11/2022

Monetary Policy
STATEMENT

Reserve Bank
of New Zealand
Te Pūtea Matua

STATEMENT
of the MPC's monetary policy strategy

The Monetary Policy Committee's (MPC's) monetary policy strategy is its overarching plan for how it will formulate monetary policy under different circumstances to achieve its objectives.[1] It outlines a consistent approach to how the MPC intends to achieve its objectives across time, accounting for trade-offs and uncertainty. Agreeing on and publishing a strategy promotes transparency, public understanding, and accountability.

Monetary policy framework and objectives

Under the *Reserve Bank of New Zealand Act 1989* (the Act), the MPC is responsible for formulating monetary policy to maintain a stable general level of prices over the medium term and to support maximum sustainable employment.[2] Operational objectives for monetary policy are set out in the Remit. The current *Remit* sets out a flexible inflation targeting regime, under which the MPC must set policy to:

- keep future annual inflation between 1 and 3 percent over the medium term, with a focus on keeping future inflation near the 2 percent mid-point; and
- support maximum sustainable employment, considering a broad range of labour market indicators and taking into account that maximum sustainable employment is largely determined by non-monetary factors.

In pursuing these objectives, the *Remit* requires the MPC to have regard to the efficiency and soundness of the financial system, seek to avoid unnecessary instability in the economy and financial markets, and discount events that have only transitory effects on inflation. The MPC must also assess the effects of its monetary policy decisions on the Government's policy to support more sustainable house prices.

The Reserve Bank's flexible inflation targeting framework and the MPC's monetary policy strategy reflect the fact that:

- low and stable inflation is monetary policy's best long-run contribution to the well-being of New Zealanders;
- in the short to medium term, monetary policy can influence real variables such as employment, and hence policy trade-offs can arise; and
- monetary policy is more effective if the Bank's policy targets are credible, so policy should be formulated in a way that ensures credibility is maintained.

Key aspects of monetary policy strategy

The MPC **practises forecast targeting**, which means that it sets monetary policy such that it expects to achieve its inflation and employment goals in the medium term. In most instances the MPC aims to return inflation to the target mid-point within a one- to three-year horizon. The appropriate horizon at each policy decision will vary based on how different policy paths will contribute to maximum sustainable employment, whether price-setters' expectations are consistent with the inflation target, and other considerations such as the balance of risks to the MPC's central economic outlook.

1 For a more in-depth discussion of monetary policy strategy in New Zealand, see J. Ratcliffe and R. Kendall (2019), 'Monetary policy strategy in New Zealand', Reserve Bank of New Zealand, *Bulletin*, Vol. 82, No. 3, April.

2 These economic objectives contribute to the overall purpose of the Act, which is to promote the prosperity and well-being of New Zealanders, and contribute to a sustainable and productive economy. See monetary policy framework for more information on New Zealand's monetary policy framework, including the full text of the *Remit*.

The MPC does not attempt to return inflation and employment to target immediately, because monetary policy actions take time to transmit through the economy. Attempting to return inflation to target too quickly would result in unnecessary instability in the economy and financial markets. The 1 to 3 percent target range for inflation provides the MPC with flexibility to ensure that managing inflation variability does not come at the cost of excessive variability in the real economy. For similar reasons, the MPC does not attempt to offset events that are expected to have only transitory effects on inflation.

The MPC **takes into account both its inflation and employment objectives** when setting policy. In the long run, no trade-off exists between the MPC's objectives. In the short to medium term, there may be situations where monetary policy can move one objective closer to target only at the cost of the other, resulting in a trade-off. When a trade-off does arise, the MPC will consider outcomes for both objectives in setting policy. In general, if employment is projected to be below its long-run sustainable level, the MPC would let inflation overshoot the target mid-point for a time, and vice versa (while staying within the 1 to 3 percent target range).

The MPC **responds to both deviations above target and deviations below target**. The MPC sets policy to stabilise employment near its maximum sustainable level, and to return inflation near to the target mid-point, regardless of whether inflation is currently below or above 2 percent. This approach helps to anchor inflation expectations at the target mid-point and promotes sustainable growth and employment by dampening fluctuations in the business cycle.

The MPC **considers the balance of risks** to its objectives that arise from uncertainty about the economic outlook and the transmission of its policy decisions. In general, the MPC will incorporate likely future developments into its central economic projections and set monetary policy in response. However, the MPC will also take into account risks to its central projections when setting policy. Under extreme uncertainty, the MPC may choose to publish scenarios instead of central projections to illustrate the range of possible situations and economic outcomes that could occur when circumstances are highly unpredictable.

The MPC **has regard to the efficiency and soundness of the financial system**, while recognising that in most instances prudential policy is better suited to leaning against risks to financial stability. The Reserve Bank takes prudential policy settings into account when setting monetary policy, and vice versa.

Implementation of strategy

The MPC applies the following process when formulating a policy decision:

1. Firstly, it assesses the outlook for the economy and the implications for its policy objectives. It then discusses risks to achieving its policy objectives.
2. Next, it considers which stance of monetary policy is most consistent with its monetary policy strategy given the current economic outlook, risks, and trade-offs.
3. Finally, the MPC decides how it will achieve the desired stance of monetary policy, including whether or not to adjust its policy settings at the current meeting and how it will communicate the policy outlook. The MPC has a suite of monetary tools to achieve its goals, and uses its Principles for Monetary Tools to make decisions on which tools to deploy.

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Thank you to RBNZ staff for photography in this report.
Cover image: Roy's Peak. Photo: Paul Conway

11/2022

Monetary Policy STATEMENT

Contents

The projection was finalised on 17 November 2022. The Official Cash Rate (OCR) projection incorporates an outlook for monetary policy that is consistent with the MPC's monetary policy assessment, which was finalised on 23 November 2022.

Policy assessment

CHAPTER 01

Castle Hill. Photo: Rebecca Williams

CHAPTER 1
Policy assessment

Tēnā koutou katoa, welcome all.

The Monetary Policy Committee today increased the Official Cash Rate (OCR) from 3.5 percent to 4.25 percent.

The Committee agreed that the OCR needs to reach a higher level, and sooner than previously indicated, to ensure inflation returns to within its target range over the medium term. Core consumer price inflation is too high, employment is beyond its maximum sustainable level, and near-term inflation expectations have risen.

Global consumer price inflation is broad based and remains heightened. Food and energy prices, and persistent core inflation, have combined to create very high headline inflation in many countries. Central banks are tightening monetary conditions in an effort to slow spending and reduce inflation pressure. The ongoing slowdown in global growth will affect New Zealand through both financial and trade channels, and impact on people's confidence due to uncertainty.

In New Zealand, household spending remains resilient, especially considering the rise in debt servicing costs, the fall in house prices, and low levels of consumer confidence. Employment levels are high, and income growth and household savings are supporting spending. The rebound in tourism is also supporting domestic demand.

The productive capacity of the economy is being constrained by broad-based labour shortages, and wage pressures are evident. Aggregate demand continues to outstrip New Zealand's capacity to supply goods and services, with a range of indicators continuing to signify broad-based inflation pressure.

Committee members agreed that monetary conditions needed to continue to tighten further, so as to be confident there is sufficient restraint on spending to bring inflation back within its 1-3 percent per annum target range. The Committee remains resolute in achieving the Monetary Policy *Remit*.

Meitaki, thanks.

Adrian Orr
Governor

SUMMARY
Record of meeting

The Monetary Policy Committee discussed developments affecting the outlook for inflation and employment in New Zealand. Inflation is currently too high and employment is beyond its maximum sustainable level. The Committee agreed it must continue to act decisively to return inflation to target and to fulfil its *Remit*.

The Committee discussed recent international economic developments. In many countries, elevated food and energy prices are contributing to high headline inflation, with high core inflation reflecting more broad based inflationary pressures. Most central banks have continued to tighten monetary conditions and to signal further interest rate increases in coming months. Financial market volatility remains high as central banks act to stem the rise in inflation in an environment of slowing and uncertain economic growth.

Expectations for global economic growth have declined further. For example, China's economy is facing headwinds emanating from the property sector, while measures to contain the spread of COVID-19 continue to cause production bottlenecks. The United States and Europe are, to varying degrees, experiencing the effects of high inflation, tighter financial conditions and associated economic uncertainty. The Committee agreed that the anticipated global growth slowdown will affect New Zealand through trade and financial channels, and increased economic uncertainty impacting on people's confidence.

The Committee observed that consumer price inflation in New Zealand in the September quarter was significantly stronger than expected. Measures of core inflation continued to rise and price pressures broadened. Survey measures and other high frequency data suggest that pricing pressure will be sustained over coming months. In addition, shorter-term inflation expectations have increased as high inflation persists.

The Committee agreed that to achieve its *Remit* objectives, actual and expected inflation need to decline substantially. Members highlighted that the longer actual inflation remains above the target band, the more likely it is that higher inflation expectations become embedded.

The Committee discussed the labour market at length, given its importance in the current economic environment. Labour shortages remain a significant constraint on economic activity. Recent data highlight a material increase in employment, enabled by a strong lift in labour force participation to a record level. Measures of labour force utilisation are near record levels and firms continue to report severe difficulties finding labour.

High consumer price index (CPI) inflation and competition for workers are putting upward pressure on wages. The Committee observed that overall wage growth is not exceeding CPI inflation after accounting for productivity growth. While wage growth for people in the same job is generally not keeping pace with inflation, many people are changing jobs or increasing their hours worked to achieve real income growth. The Committee noted that public sector wage growth has lagged that in the private sector and agreed this lag represents an upside risk to wage pressure going forward.

The Committee observed the stronger than expected rebound in tourism since New Zealand's border reopened. Short-term visitor arrivals, international card spending data, and information gathered from recent business visits indicate that tourism spending will make a strong contribution to economic activity in coming months. However, some members noted that ongoing capacity constraints could, at some point, inhibit the tourism recovery and add to overall inflation pressures. In relation to New Zealand's goods exports, members observed that a lower New Zealand dollar is currently mitigating the impact of recent declines in international commodity prices.

The Committee discussed the recent *Financial Stability Report* and noted that the financial system remains resilient. In particular, members highlighted that recent stress tests demonstrate banks' resilience. The Committee was especially interested in the resilience of household balance sheets to scenarios of higher interest rates, reduced labour demand, and declining house prices. The Committee noted that while national house price indices have declined to mid-2021 levels, they are still above estimates of sustainable house prices and above levels that prevailed pre-COVID-19.

Household spending remains robust, especially considering the rise in debt servicing costs, the fall in house prices, and low levels of consumer confidence. Members observed that the large stock of household savings, in addition to income growth, may provide a buffer to support consumption now and in the future. However, members also noted that the ownership of savings is likely concentrated, leaving the majority of households exposed to high inflation and interest rates.

The Committee agreed that as debt servicing costs rise, spending decisions for many households will be increasingly constrained. These constraints would be most felt by recent home buyers with a high debt servicing commitment relative to their income. The Committee agreed that the impact of rising interest rates on households' spending and saving decisions is an important channel for monetary policy. In addition to constraining spending, higher interest rates also encourage saving and paying down of debt.

The Committee discussed domestic financial conditions, noting that wholesale interest rates have risen significantly since the August *Statement*, primarily due to higher-than-expected inflation in New Zealand and globally. Retail lending rates have also increased but remain lower than the levels wholesale rates might imply. Members observed that this reflects a combination of both the higher volume and the mix of current bank funding. Members noted that a gradual normalisation in bank funding conditions over the forecast period could result in sustained upward pressure on retail lending rates.

The Committee considered the economic projections. Members noted that a reduction in aggregate demand is projected to cause GDP in the New Zealand economy to temporarily contract by around 1% from 2023. Members noted that this reduction in aggregate demand was necessary to return inflation to target over the forecast period. Members agreed that the exact timing and extent of negative GDP growth was difficult to predict, but historical evidence suggests risks are skewed toward a likely short period of contraction. Members also agreed that the sooner supply and demand were better matched in the economy, the lower the overall cost of reducing inflation.

Members agreed that monetary policy primarily impacts on demand in the economy. However, any increase in the supply potential of the economy, such as through productivity improvements, would also assist in reducing inflation. However, members agreed that a significant increase in the economy's capacity to supply goods and services could not be relied on to reduce inflation pressures over the forecast horizon.

The Committee agreed that fiscal policy can also act to reduce demand in the economy. Members observed that in a higher inflation environment, a given level of government services would cost more to deliver. However, members noted that inflation would also lead to increased government revenues in nominal terms, potentially offsetting the rising cost of service delivery. On balance, members viewed the risks to inflation pressure from fiscal policies as skewed to the upside given the ongoing real demand for services.

The Committee received an update on the status of the Large Scale Asset Purchase (LSAP) portfolio and noted that sales of bonds in the LSAP portfolio to New Zealand Debt Management began in July. Members observed that the New Zealand government bond market continues to function normally under the current pace of LSAP sales and agreed to continue to evaluate this on an ongoing basis. The Committee agreed that the level of settlement cash balances is not a source of unexpected inflationary pressure and noted that overnight wholesale interest rates remain aligned to the OCR.

The Committee discussed the extent of additional monetary tightening required to achieve its *Remit*. Members agreed that the OCR needed to reach a level where the Committee could be confident it would reduce actual inflation to within the target range over the forecast horizon. Members agreed that this level had increased since the time of the August *Statement* due to the persistence of inflationary pressures resulting in a higher short-term nominal neutral interest rate.

The Committee discussed the size of the OCR increase to be delivered at this meeting. Increases of 50, 75 and 100 basis points were considered. The Committee discussed the relative merits of maintaining consistent increments in the OCR versus moving more quickly to reach the higher level of the OCR required. Members agreed that a larger increase in the OCR was appropriate, given the resilience of domestic spending, and the higher and more persistent actual and expected inflation outcomes.

The Committee gave consideration to an increase in the OCR of 75 or 100 basis points. On the balance of risks, the Committee agreed that a 75 basis point increase was appropriate at this meeting. Members highlighted that the cumulative tightening of monetary conditions delivered to date continues to pass through to the economy via the lagged transmission to effective retail interest rates.

On Wednesday 23 November, the Committee reached a consensus to increase the OCR from 3.5% to 4.25%.

Attendees:

Reserve Bank members of MPC:

Adrian Orr, Karen Silk, Christian Hawkesby, Paul Conway

External MPC members:

Bob Buckle, Peter Harris, Caroline Saunders

Treasury Observer:

Dominick Stephens

MPC Secretary:

David Craigie

Current economic assessment and monetary policy outlook

CHAPTER 02

Emerald Lakes, Tongariro National Park. Photo: Nele McHattie

CHAPTER 2
Current economic assessment and monetary policy outlook

Key points

- The economic costs of high inflation mean that the best contribution monetary policy can make to promote the prosperity and wellbeing of New Zealanders is to ensure that inflation returns to being low and stable. Higher interest rates will slow demand in order to relieve pressure on available resources and enable the Monetary Policy Committee (MPC) to sustainably meet its inflation and employment objectives over the medium term.
- Demand in the New Zealand economy has remained resilient to global and domestic challenges to date. Household spending has stayed elevated despite falling house prices, high consumer price inflation and increasing interest rates. Spending has been supported by high employment, increasing wages, and savings built up in recent years. Global prices for some of our key export commodities remain at high levels, although some have eased in recent months. The tourism sector has seen a strong recovery in international visitors, and will contribute significantly to economic growth over coming quarters.
- However, demand in the New Zealand economy is not being matched with enough supply. Business activity is being held back by severe labour shortages that are unlikely to ease in the near term. The unemployment rate remained very low at 3.3 percent in the September 2022 quarter, despite a record share of the population participating in the labour force. Although there are signs that supply-chain bottlenecks are starting to loosen globally, they continue to delay the production and delivery of some goods in New Zealand and are adding to overall costs.
- The very tight labour market is contributing to high consumers price index (CPI) inflation. Wage growth is increasing, adding to business costs while also supporting household incomes and spending. Housing construction costs remain high and continue to increase. Annual non-tradables inflation was 6.6 percent in the September 2022 quarter.
- Global developments have also been a large contributor to high headline inflation in New Zealand, with annual tradables inflation at 8.1 percent in the September 2022 quarter. Increases in commodity and energy prices as a result of the Ukraine war have compounded existing inflationary pressures that emerged during the COVID-19 pandemic. Inflation is now at or near multi-decade highs in many parts of the world.

- Central banks have responded to high inflation by raising interest rates. Higher interest rates globally have contributed to a weaker New Zealand dollar trade-weighted index (TWI) exchange rate. All else equal, a lower exchange rate supports exporters' incomes and makes the cost of imported goods and services more expensive, adding to inflation in New Zealand.
- CPI inflation remains too high. Due to the domestic and global factors outlined above, annual CPI inflation was 7.2 percent in the September 2022 quarter – well above the MPC's 1 to 3 percent target band over the medium term. Measures of persistent or 'core' inflation have also increased further. Inflation expectations, as measured by the Reserve Bank's *Survey of Expectations*, for one, two, five and ten years ahead all increased following the latest CPI data, after showing tentative signs of turning in the previous quarter. Increases have been larger at the one- and two-year horizons.
- The MPC has been tightening monetary policy since mid-2021. By increasing the official cash rate (OCR), the MPC is aiming to reduce demand in the economy so that it better aligns with the economy's ability to sustainably supply goods and services. Higher interest rates are intended to reduce household and business spending with the usual lag of several quarters.
- Higher interest rates globally are also intended to ease demand offshore as central banks try to lower inflation in their own economies. This will eventually lead to lower imported inflation. Lower global growth is expected to contribute to lower business investment and lower demand for our export goods over coming years.
- Conditional on our central economic outlook, it is expected that the OCR will need to increase more than assumed in the August *Statement*. The higher outlook for interest rates reflects stronger overall inflationary pressures, including those resulting from the recent increases in inflation expectations. This more than offsets the medium-term effects of weaker global growth on the New Zealand economy.
- Because the New Zealand economy is starting from a position of very high inflation and acute labour shortages, a period of economic contraction is likely as economic activity declines from elevated levels. The peak to trough decline in the level of gross domestic product (GDP) is expected to be about 1 percent. Employment may fall below its maximum sustainable level for a time.
- Trying to avoid an economic contraction by limiting any interest rate increases in the near term would likely lead to a longer period of high inflation. In turn, this would likely result in higher interest rates and a larger contraction eventually being required to bring inflation and employment back to a more sustainable path.

Current economic assessment

Demand remains resilient

Demand in the New Zealand economy has shown continued resilience to global and domestic headwinds. GDP increased 1.7 percent in the June 2022 quarter, and recent data suggest that growth has remained firm over the second half of this year (figure 2.1).

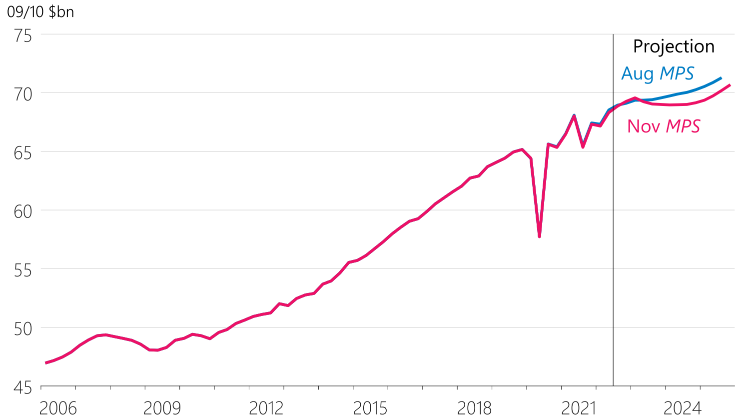
Household spending has remained elevated, despite high inflation, rising interest rates, falling house and other asset prices, and increased global uncertainty. Although private consumption declined 3.1 percent in real terms in the June 2022 quarter, this in large part reflected changing seasonal patterns since the onset of the COVID-19 pandemic. Real consumption per person remains above pre-pandemic levels (figure 2.2), and higher-frequency spending data indicate that this strength has continued during the second half of this year.

Recent household spending has been supported by rising incomes, due to the combination of high employment, increasing wages, and the Government's Cost of Living Payment. Savings built up by the household sector over recent years – most notably during lockdowns when spending opportunities were limited – have also strengthened household balance sheets in aggregate (figure 2.3).

Figure 2.1

Quarterly production GDP

(real, seasonally adjusted)

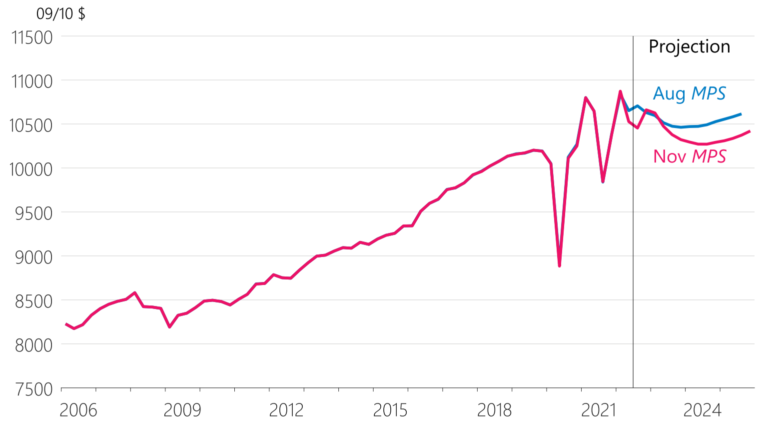


Source: Stats NZ, RBNZ estimates.

Figure 2.2

Quarterly private consumption expenditure per working-age person

(real, seasonally adjusted)

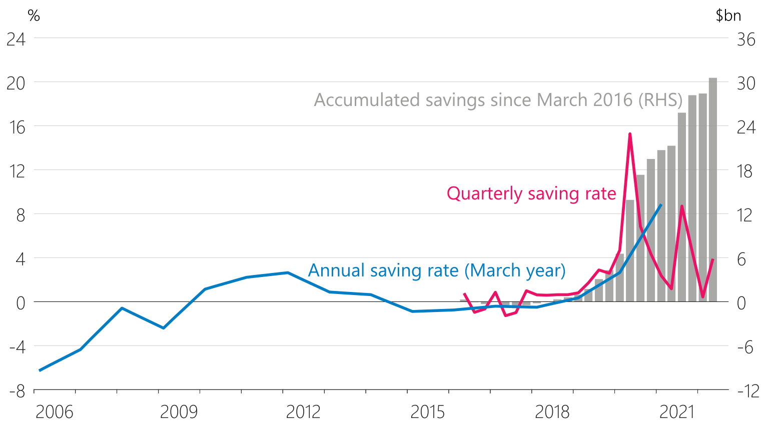


Source: Stats NZ, RBNZ estimates.

Figure 2.3

Saving rate as a share of net disposable income and accumulated savings

(seasonally adjusted)



Source: Stats NZ, RBNZ estimates.

Although New Zealand house prices have now declined by about 11 percent since the November 2021 peak, household wealth in the June 2022 quarter was still 15 percent higher than at the end of 2020. The effect of the most recent reduction in household wealth on spending has also been outweighed by higher incomes so far. Although the outlook for residential construction has deteriorated due to falling house prices and higher interest rates, the number of new residential consents has yet to decline significantly, and forward order books for many construction firms remain full.

The recovery in international visitors has been faster than anticipated since New Zealand's borders were reopened (figure 2.4). The increase in visitors from Australia has been particularly large. Businesses that Reserve Bank staff have spoken to during the past six weeks have said that forward bookings are also strong, with increasing bookings being made by traditionally higher-spending travellers from the US, Canada and the UK. The recovery in the tourism sector is expected to make a large contribution to economic growth over New Zealand's upcoming summer.

World prices for our commodity exports remain at elevated levels, despite a weakening global outlook. High prices are supporting exporters' incomes and businesses that serve farmers in particular.

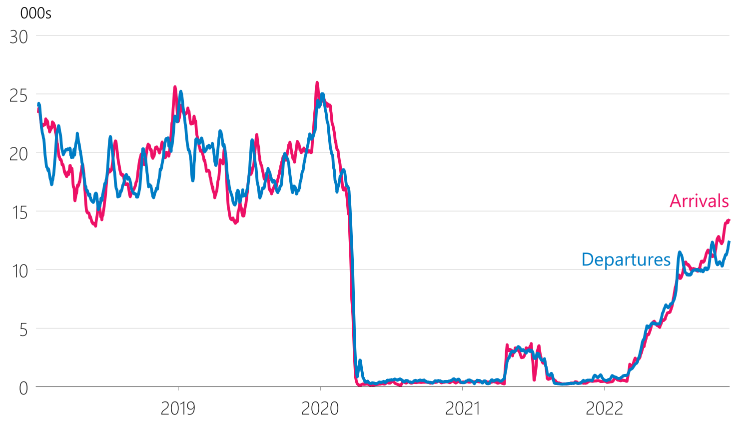
Labour shortages are constraining output

Although demand has been resilient to date, it is not being matched with sufficient supply. Acute labour shortages are limiting business services and output across many industries, regions and skill types (figure 2.5). The unemployment rate remained low at 3.3 percent in the September 2022 quarter (figure 2.6), despite a record number of New Zealanders opting to participate in the labour force.

Figure 2.4

Air passenger arrivals and departures

(7-day moving average)

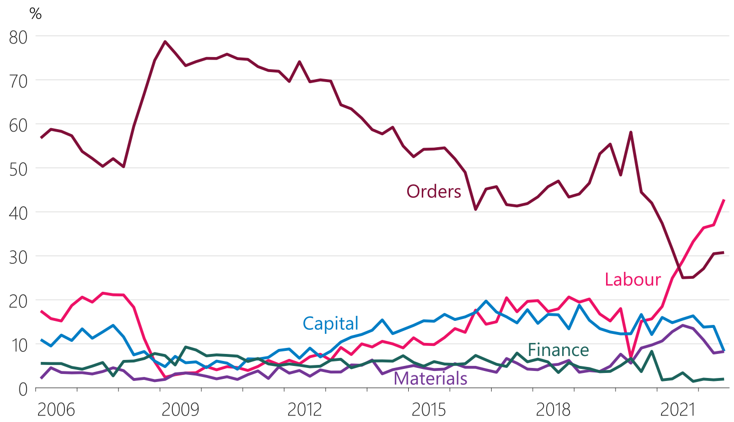


Source: New Zealand Customs Service.

Figure 2.5

QSBO factor most limiting production

(seasonally adjusted)

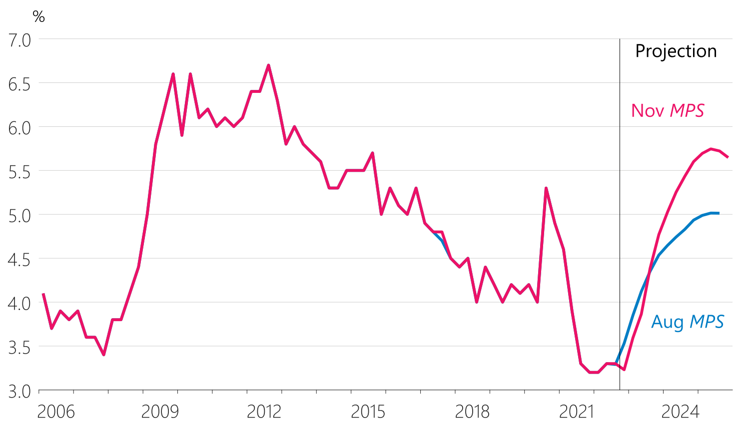


Source: NZIER, RBNZ estimates.

Figure 2.6

Unemployment rate

(seasonally adjusted, unemployed as share of the labour force)



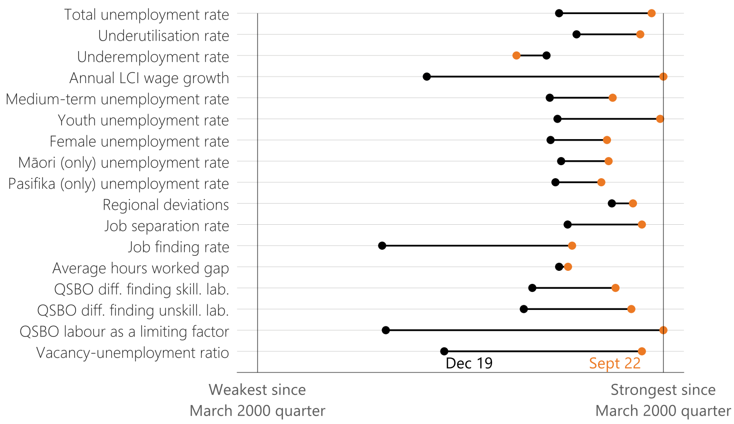
Source: Stats NZ, RBNZ estimates.

The low unemployment rate is one of many indicators that suggest employment is currently above its maximum sustainable level (figure 2.7). The unemployment rate is assumed to stay low in the near term, as employers are likely to hold on to existing workers and instead reduce vacancies in response to the weakening outlook for economic growth.

Labour shortages are providing employees with a greater ability to demand higher wages in response to increased living costs. As discussed further in section 4.2, although same-job wage inflation has not kept pace with higher CPI inflation, broader measures of average hourly earnings and overall incomes have increased at a greater pace (figure 2.8). The tight labour market has seen workers being promoted, switching jobs or increasing their hours to boost household incomes, helping sustain continued growth in household spending.

Figure 2.7

Maximum sustainable employment (MSE) indicator suite



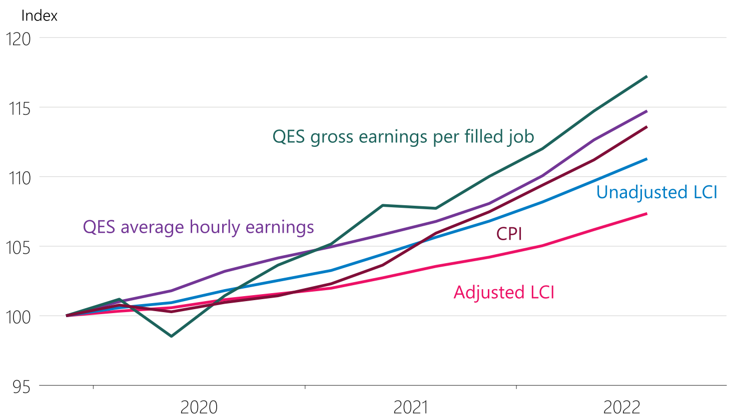
Source: NZIER, Stats NZ, MBIE, ANZ, RBNZ estimates.

Note: The Reserve Bank uses a range of indicators when assessing MSE. It regarded the December 2019 quarter as a period when this selection of indicators pointed to employment being at MSE. However, current outturns should not be directly compared to the December 2019 quarter as the level of MSE changes over time. The vertical lines on the left-hand and right-hand sides represent the lowest and highest data outcomes since 2000, respectively. An orange dot to the right of a black dot means that the latest data outcome was stronger than in the December 2019 quarter.

Figure 2.8

Wage and labour income compared to CPI

(index=100 in 2019Q4)



Source: Stats NZ.

Note: The adjusted LCI (all sectors, salary and ordinary wage rates) estimates wage growth for the same job, being done to the same quality and quantity. The unadjusted LCI is the same as the adjusted LCI but includes wage growth due to changes in what quality a job is done to. QES average hourly earnings (all industries, ordinary and overtime, seasonally adjusted) is an hourly labour income measure that reflects same-job wage growth but also income changes due to productivity changes, job promotions and job switching. QES gross earnings (all industries, seasonally adjusted) per QES filled job (all industries, seasonally adjusted) includes all these factors as well as changes in the number of hours worked. For more information, see table 4.1.

While labour is widely reported as the biggest factor holding businesses back from expanding production, materials shortages and supply-chain bottlenecks also continue to constrain production and increase overall costs. There have been some signs of supply-chain bottlenecks loosening in parts of the world, but this has yet to translate meaningfully into improved supply conditions or easing cost pressure for New Zealand firms (figure 2.9).

The tight labour market is contributing to inflation

Pressure on available resources and increasing wages are contributing to high domestic inflation. Annual non-tradables inflation – which captures price changes for goods and services that are relatively less exposed to international competition – was 6.6 percent in the September 2022 quarter, up from 6.3 percent in the June quarter. High non-tradables inflation continues to be accounted for in large part by construction, rent and other housing-related price growth (figure 2.10). However, inflation has become even more broad-based as businesses have passed on higher wage and general costs into consumer prices.

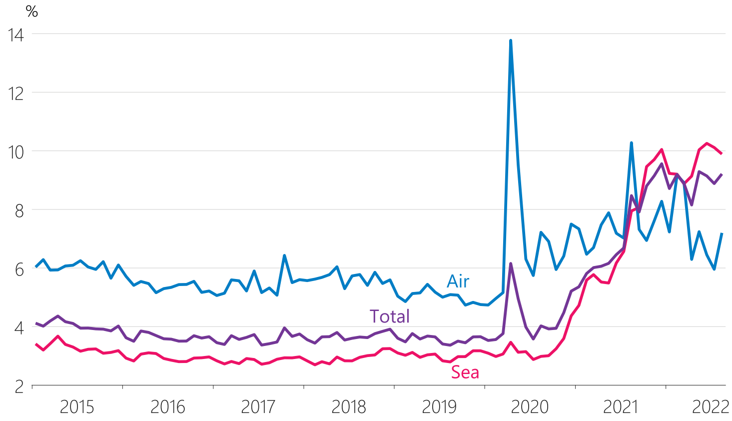
Global developments are also adding to inflation in New Zealand

Global developments have also been a large contributor to high headline inflation in New Zealand. Annual tradables inflation was 8.1 percent in the September 2022 quarter. While this was lower than the 8.7 percent inflation in the year to the June 2022 quarter, it is in stark contrast to the decade of low or falling tradables prices before the COVID-19 pandemic and was higher than expected at the time of the August *Statement*. Increases in food, other commodity and energy prices as a result of the Ukraine war compounded existing inflationary pressure that had emerged during the COVID-19 pandemic. Inflation is now at or near multi-decade highs in many parts of the world.

Figure 2.9

Shipping costs for consumer goods

(share of import value for duty)



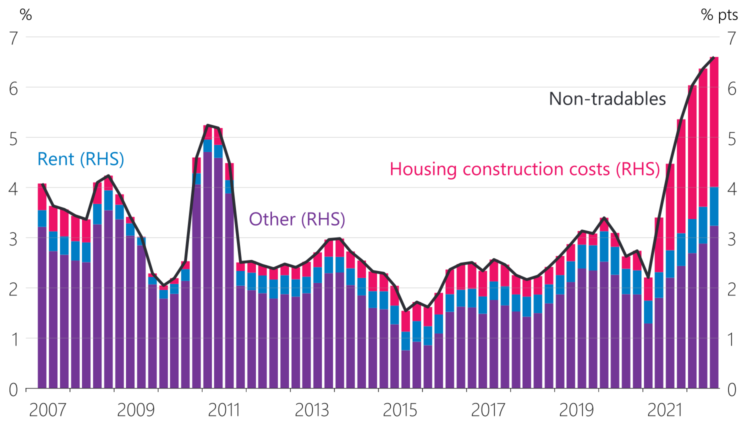
Source: Stats NZ, RBNZ estimates.

Note: These series are estimated by taking the value of consumer merchandise imports including freight and insurance costs (CIF) and subtracting the reported value for duty (VFD), which excludes these costs. They are expressed as a percentage of the VFD figure.

Figure 2.10

Contributions to non-tradables inflation

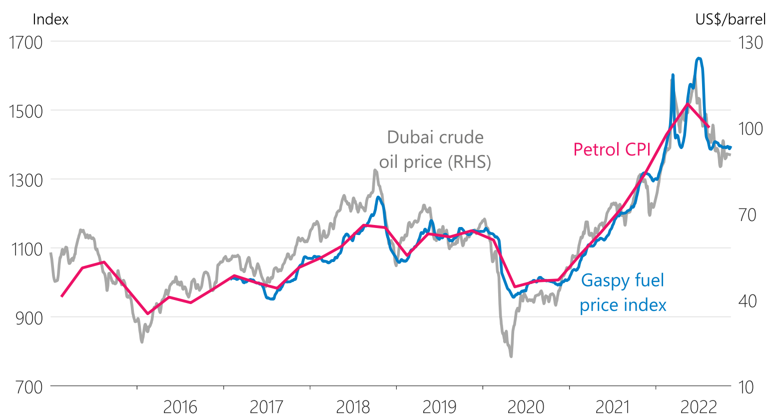
(annual)



Source: Stats NZ.

Figure 2.11

Petrol and Dubai oil prices



Source: Datamine – Gaspy data, Stats NZ, Reuters, RBNZ estimates.

Note: Gaspy fuel prices are indexed to the petrol CPI level in the December 2019 quarter.

Although global oil prices have eased from their March 2022 peak, they remain 14 percent higher than they were at the start of 2022, contributing to high petrol prices (figure 2.11). Food and fertiliser prices remain high globally. In conjunction with domestic weather events and increasing wage costs, these global factors have contributed to high food price inflation in New Zealand (figure 2.12). More generally, high inflation globally has also led to sharp increases in world prices for the goods that New Zealand imports.

Central banks around the world are responding to high inflationary pressures by increasing interest rates – rapidly in some cases. High interest rates and elevated geopolitical uncertainty, such as that related to the war in Ukraine, have seen the outlook for global growth continually revised lower over the past year. Interest rate increases in many of our trading partner economies, especially the US, have put downward pressure on the New Zealand dollar TWI exchange rate. Global political and economic events have also reduced financial market participants' tolerance for risk since the August *Statement*, strengthening demand for US dollar denominated assets and contributing to a lower New Zealand dollar. A lower TWI supports exporters' returns and makes New Zealand's imports more expensive, adding to inflation in New Zealand.

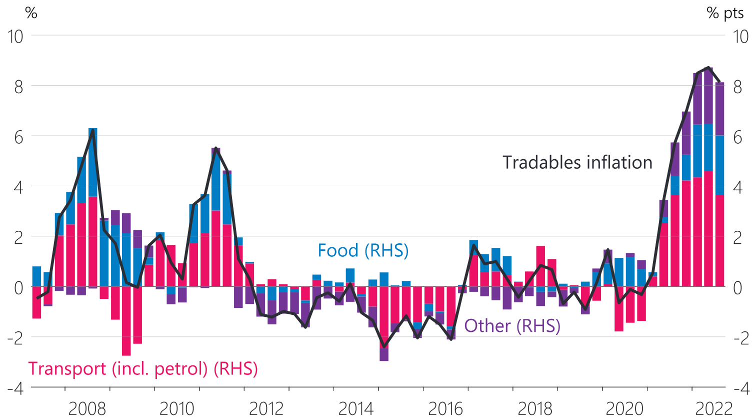
Core and expected inflation have increased

Due to these global and domestic factors, annual CPI inflation was 7.2 percent in the September 2022 quarter (figure 2.13). This was a very slight decline from the 7.3 percent annual inflation in the previous quarter, but remains well above the MPC's 1 to 3 percent target band over the medium term. Consistent with inflation becoming more broad-based (figure 2.14), measures of persistent or 'core' inflation have also increased further (figure 2.15).

Figure 2.12

Contributions to tradables inflation

(annual)

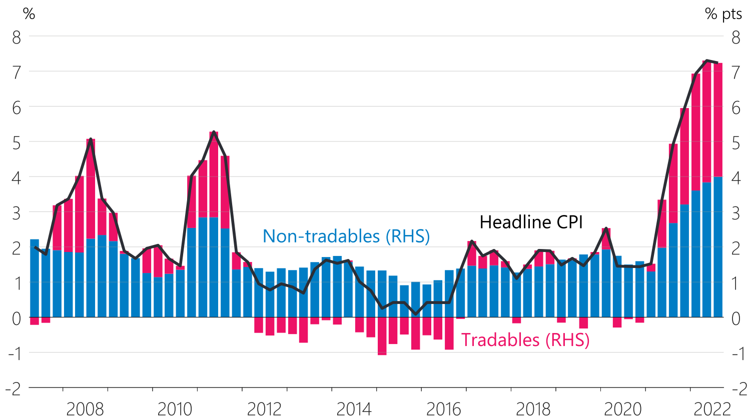


Source: Stats NZ.

Figure 2.13

Contributions to CPI inflation

(annual)

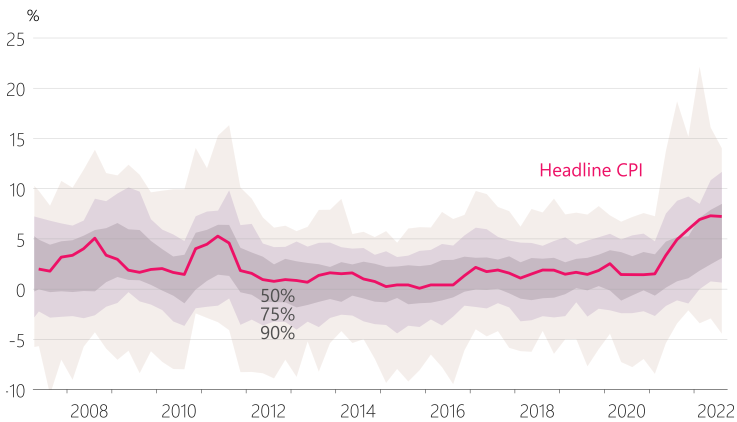


Source: Stats NZ.

Figure 2.14

Distribution of price movements in CPI classes

(annual)



Source: Stats NZ.

Note: There are more than 100 CPI classes – capturing prices for specific types of goods and services – for which the bands show the distribution of annual percent changes. Shaded areas represent: grey – middle 50 percent of CPI classes; purple – middle 75 percent of CPI classes; light brown – middle 90 percent of CPI classes.

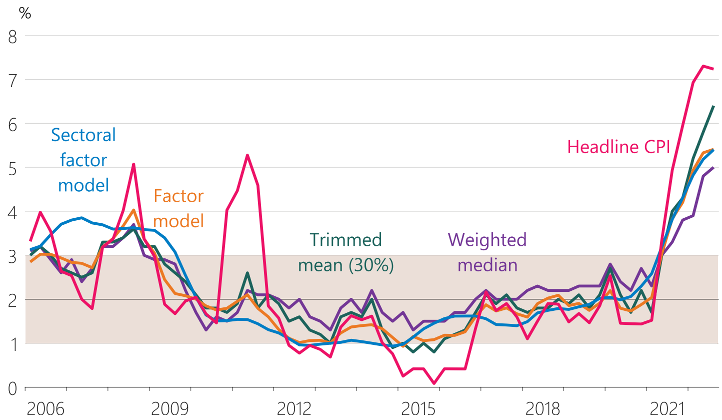
Expectations for inflation one, two, five and ten years ahead have also increased following the release of the latest CPI data (figure 2.16). Expectations for inflation one and two years ahead have increased by 0.2 and 0.5 percentage points respectively. Five and ten year expectations have increased by 0.1 percentage points. Increases in inflation expectations imply that the short-term nominal neutral rate has increased, meaning monetary policy has not been as contractionary as previously assumed (see section 4.1).

Higher core and expected inflation suggests that inflation will remain elevated in the near term and will only start to decline significantly in the second half of 2023. The announced reintroduction of fuel-related taxes in January 2023 and the expiry of transport subsidies will result in a one-off boost to inflation in the March 2023 quarter.

Higher domestic interest rates will slow the pace of economic growth and reduce inflation

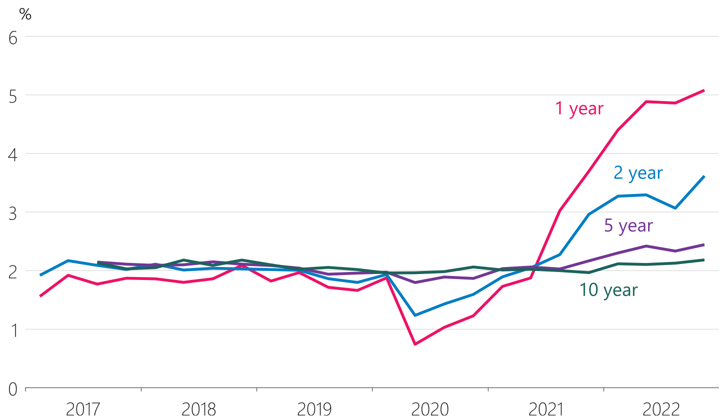
The MPC began tightening monetary policy in mid-2021 in response to increasing inflationary pressures and a tight labour market. Actual and expected OCR increases have seen the interest rates faced by households and businesses increase quickly over the past 12 months. Higher interest rates – in conjunction with low population growth, strong building activity and regulatory changes – have already contributed to a decline in house prices from their November 2021 peak. Although the speed and extent of the decline is highly uncertain, house prices are assumed to continue to fall towards more sustainable levels over the coming year (figure 2.17). The central projection assumes that house prices will decline by 20 percent in total from the November 2021 peak.

Figure 2.15
Headline and core inflation measures
(annual)



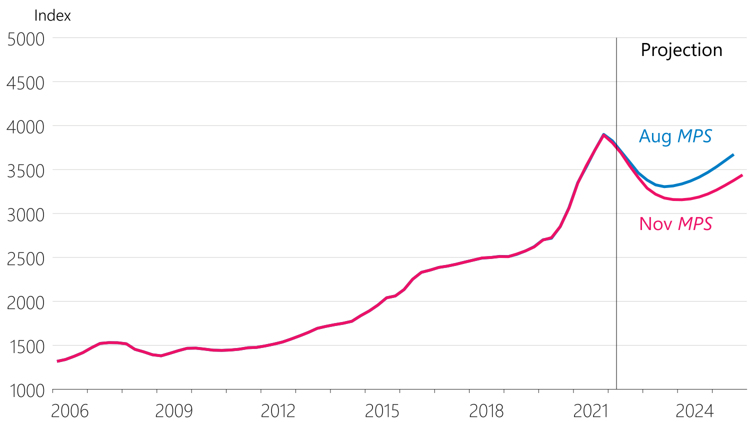
Source: Stats NZ, RBNZ estimates.

Figure 2.16
Inflation expectations
(annual, years ahead)



Source: RBNZ *Survey of Expectations* (Business).

Figure 2.17
House prices
(nominal)



Source: CoreLogic, RBNZ estimates.

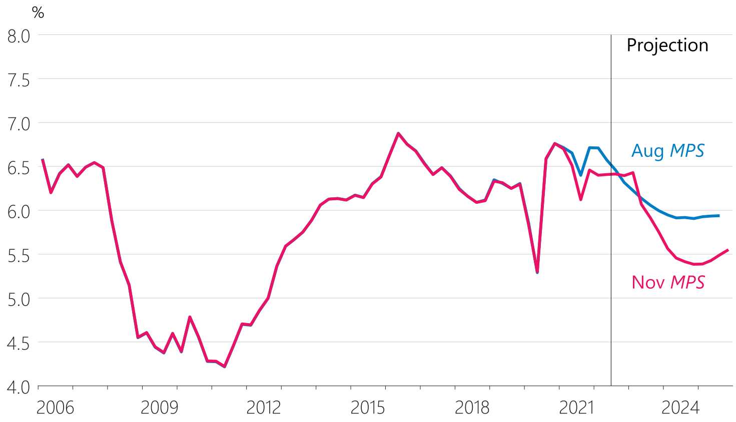
Higher interest rates, elevated construction costs and lower house prices are also expected to result in a decline in residential building from next year. Residential investment is assumed to remain elevated in the near term as builders work through the large pipeline of consented projects, before declining significantly as a share of the economy over the medium term (figure 2.18).

Lower house prices and higher interest rates are assumed to result in a sustained slowing in household spending, as households feel less wealthy and a greater portion of their income is directed towards servicing debt. Although household spending is currently supported by robust income growth, employment growth is expected to wane as higher interest rates contribute to lower overall spending and reduce businesses' demand for workers. Consumption per capita is assumed to decline over coming years (figure 2.2).

The weaker global outlook will lower economic growth and inflation in New Zealand over coming years

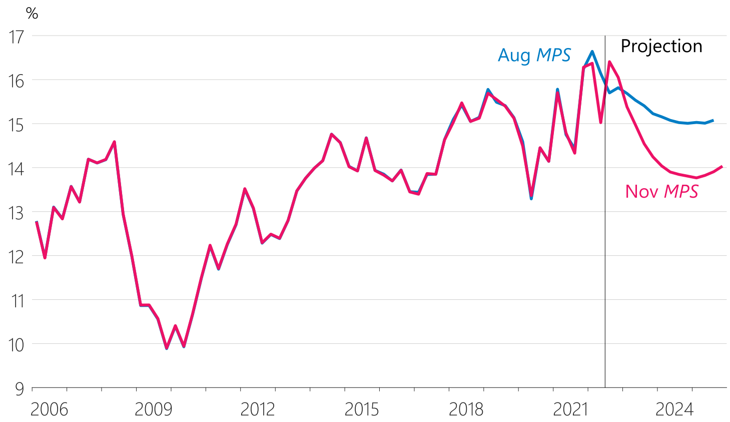
Higher interest rates globally are intended to ease demand as central banks try to lower inflation in their own economies. While higher interest rates abroad are currently placing downward pressure on the New Zealand dollar exchange rate and boosting prices for our imports, slowing global growth should eventually lead to lower imported inflation over the medium term. Lower global demand and elevated uncertainty are expected to contribute to both lower business investment and demand for our export goods over coming years (figures 2.19 and 2.20).

Figure 2.18
Residential investment
(real, share of potential GDP, seasonally adjusted)



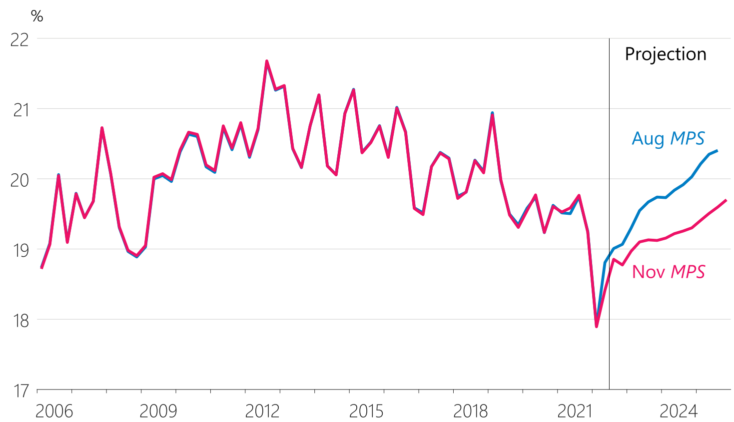
Source: Stats NZ, RBNZ estimates.

Figure 2.19
Business investment
(real, share of potential GDP, seasonally adjusted)



Source: Stats NZ, RBNZ estimates.

Figure 2.20
Exports of goods
(real, share of potential GDP, seasonally adjusted)



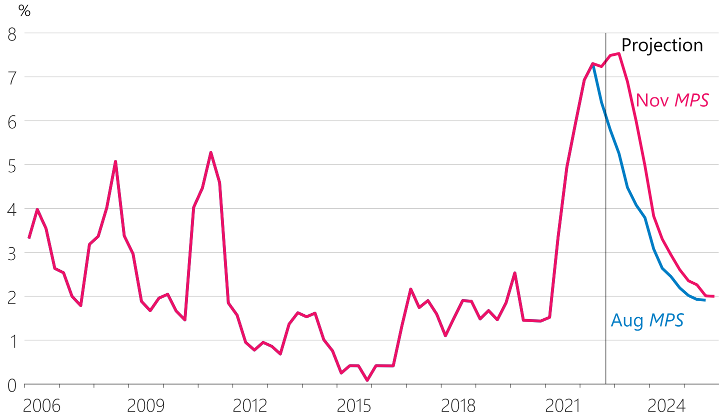
Source: Stats NZ, RBNZ estimates.

Annual inflation will return to 2 percent over the medium term

Non-tradables and tradables inflation are expected to ease over the medium term, following some near-term strength. Lower demand will result in less pressure on the available supply of labour and goods, which means less pressure on prices and lower domestic inflation. Higher interest rates globally are assumed to lead to a simultaneous slowing of growth in our trading partner economies, which will result in easing imported inflationary pressures and demand for New Zealand's goods exports. Annual CPI inflation is assumed to return to within the 1 to 3 percent target band in the second half of 2024, and to 2 percent by the end of the projection (figure 2.21).

Because New Zealand is starting from a position of very high inflation and acute labour shortages, an economic contraction is likely. This is discussed further in Box A.

Figure 2.21
CPI inflation
(annual)



Source: Stats NZ, RBNZ estimates.

Box A

Economic contraction in New Zealand

The New Zealand economy – like many economies around the world – is expected to enter recession in 2023. This economic contraction will, in part, be a result of higher interest rates, as the Reserve Bank acts to reduce inflation and return employment to a more sustainable level. It will therefore differ from contractions experienced in New Zealand in recent decades when poor economic outlooks, often related to international conditions such as those during the global financial crisis, were cushioned by lower interest rates.

Government spending and monetary policy measures around the world supported households and businesses from the onset of the COVID-19 pandemic. When COVID-19 restrictions were first eased globally, the combination of strong pent-up demand, labour shortages and bottlenecks in production and trade led to higher inflation. Just as it was becoming increasingly clear that inflation was going to remain higher for longer than expected, the Russian invasion of Ukraine in February 2022 led to higher energy and food prices. Annual inflation has now reached 30-year highs in many economies around the world.

In New Zealand, annual inflation was 7.2 percent in the September 2022 quarter. High and variable inflation makes it harder for households and businesses to plan and to know where to best direct any investment. Unexpected changes in inflation also arbitrarily affect relative outcomes for savers and borrowers. The higher cost of living is particularly challenging for people earning low or fixed incomes, as it erodes purchasing power and is effectively a pay cut. Given the economic costs of high inflation, the best contribution that monetary policy can make to promote the wellbeing of New Zealanders and support maximum sustainable employment (MSE) is to ensure inflation returns to being low and stable.

MSE is unobservable and cannot be represented by a single measure, such as the unemployment rate, but it captures an environment where the balance of supply and demand in the labour market is not causing consumer prices and wages to move in an unsustainable way. If wages are increasing faster than productivity growth, this can lead to higher and more persistent inflation as businesses pass on these costs. This can then lead to even higher wages as workers seek compensation for an even higher cost of living. When labour demand far outstrips labour supply, as is the case in New Zealand currently, the economy bears higher costs. These include rising inflation and ongoing disruptions.

To return inflation to the target range and employment to a more sustainable level, the MPC needs to increase interest rates so overall spending declines from recent high levels. This will slow demand for output and for workers. In turn, this will reduce pressure on the supply capacity of the economy, leading to less pressure on costs, prices and inflation. The easing in global growth from higher interest rates overseas will also contribute to lower domestic activity and inflation, by slowing demand for our exports and reducing the global prices of our imports respectively.

Because the New Zealand economy is starting from a position of very high inflation and acute labour shortages, an economic contraction is likely. Any fiscal or monetary policy actions that try to offset this with further stimulus will ultimately lead to a longer period of high inflation, with higher interest rates subsequently needed to control inflation and return the economy to a more sustainable path. The peak to trough decline in the level of GDP is expected to be about 1 percent. In the central projection, this recession is assumed to be spread over several quarters, although there is uncertainty about the timing.

A consequence of slowing economic growth is that the unemployment rate is likely to increase. It is possible that labour market conditions fall below their maximum sustainable level for a time. Keeping unemployment near current levels without causing unsustainable wage pressures would require structural changes in the labour market that are beyond the control of the Reserve Bank.

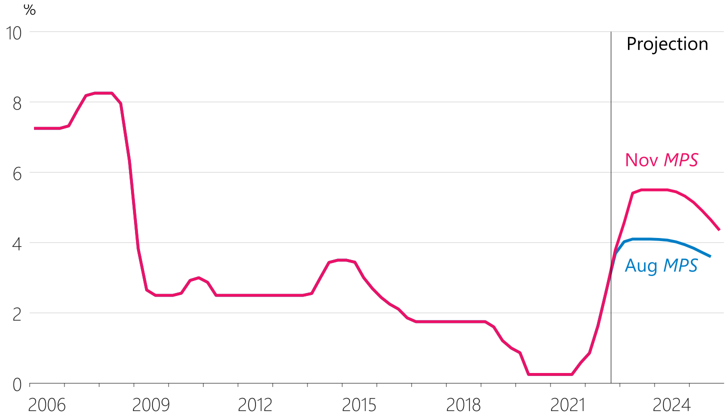
Higher interest rates will reduce aggregate demand and consequently pressure on labour and other production resources, such as materials and machinery. Another way to relieve this pressure would be for the supply capacity of the economy to grow – for example, through increased productivity here or abroad. If this were to occur, it would mean that a smaller economic contraction would be consistent with lower inflation and maximum sustainable employment. However, it is highly unlikely that such an adjustment will occur soon enough to address current inflationary pressures.

Monetary policy outlook

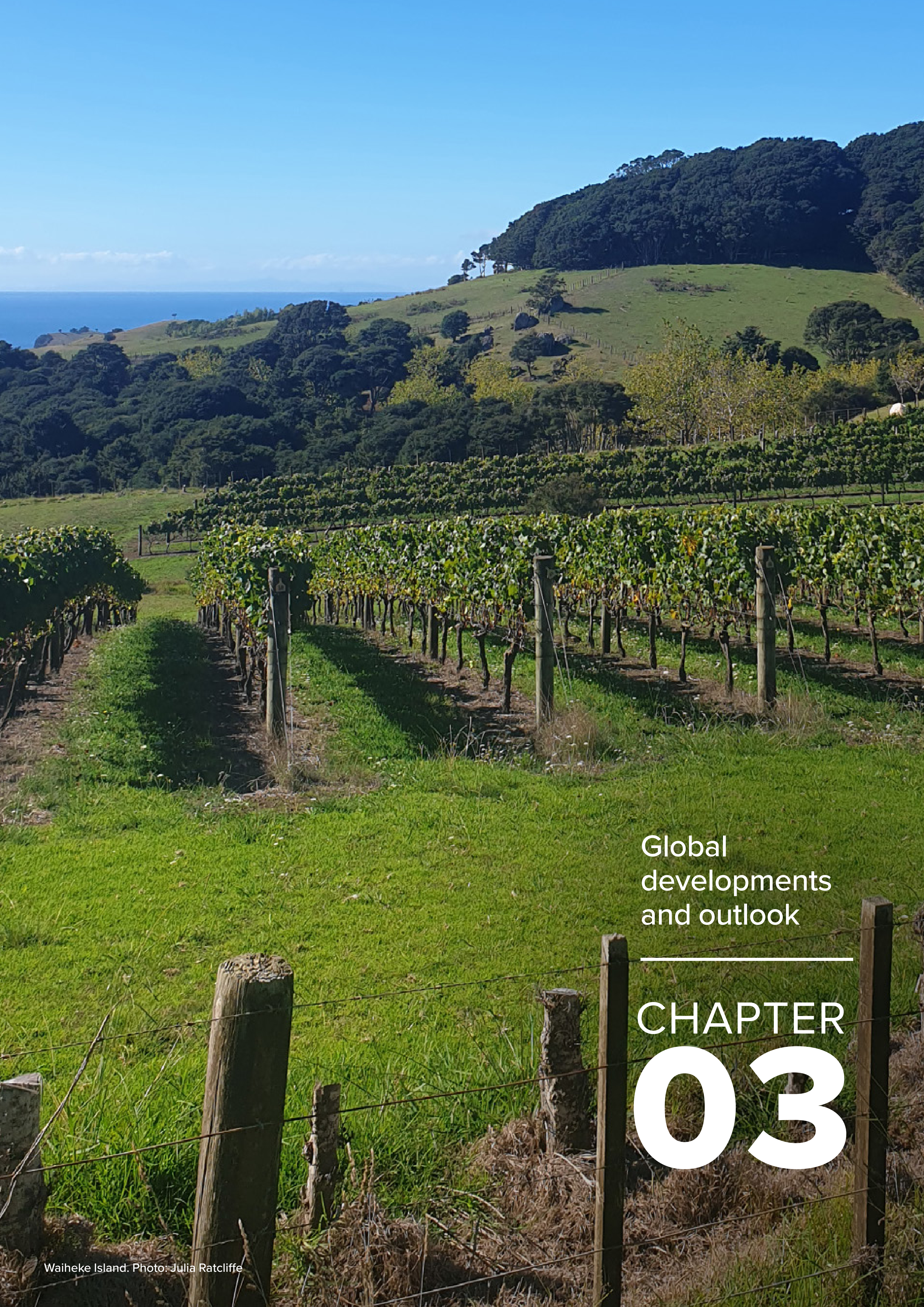
Inflation is starting from a higher level than assumed in the August *Statement*, and inflationary pressures are also assumed to be higher over the medium term. Continued labour shortages, recent increases in inflation expectations and a stronger recovery in international tourism are expected to underpin momentum in domestic inflation. Inflationary pressures are also stronger globally. Higher global interest rates in response to these inflationary pressures are placing downward pressure on New Zealand's exchange rate, which – all else equal – results in a higher near-term outlook for imported inflation.

The magnitude, speed, and widespread nature of interest rate increases globally have seen the outlook for global growth deteriorate, and increased the risk that this slowdown will be more severe than intended. However, the assumed impacts of weaker global activity are more than offset by the stronger inflation outlook within New Zealand. Conditional upon our central economic outlook, the OCR is projected to increase by more than assumed at the time of the August *Statement* (figure 2.22). As outlined in Box A, higher interest rates will ensure that demand slows in order to relieve pressure on available resources and enable the MPC to meet its inflation and employment objectives. Ensuring that inflation returns to being low and stable is the best contribution monetary policy can make to long-term economic wellbeing and maximum sustainable employment.

Figure 2.22
OCR
(quarterly average)



Source: RBNZ estimates.



Global developments and outlook

CHAPTER 03

Waiheke Island. Photo: Julia Ratcliffe

CHAPTER 3
Global developments and outlook

Weakening economic activity indicators foreshadow a decline in global growth

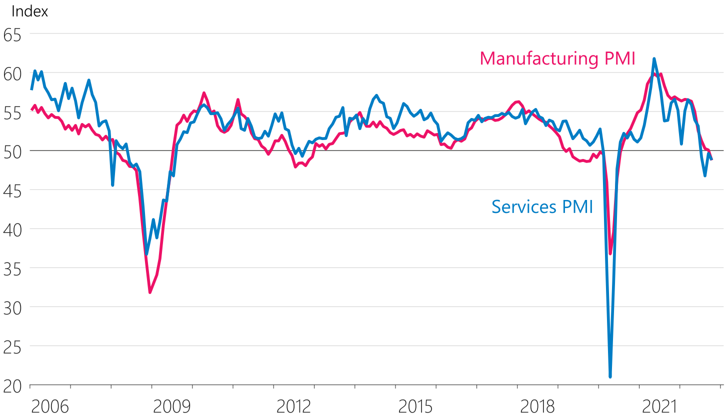
The outlook for global growth has continued to weaken since the August *Statement*. This weaker outlook reflects spillovers from the Ukraine war, tighter financial conditions, low real income growth, and a slowdown in housing activity across the globe.

GDP growth has slowed in most of New Zealand's major trading partners over 2022. Economic activity in the US and Europe has expanded only slightly since the end of 2021, while stringent COVID-19 controls have dampened growth significantly in China. In contrast, GDP growth in Australia has remained relatively firm at 3.6 percent on an annual basis.

Timely indicators of economic growth – such as Purchasing Managers Indices (PMI), a measure of manufacturing activity – have declined through 2022. Aggregate advanced economy manufacturing PMIs are just within expansionary levels, while services PMIs have been contractionary for several months (figure 3.1). Global consumer confidence measures have remained at low levels since early 2022.

Global household spending growth has moderated, but has nonetheless remained at a relatively high level. Savings built up during the COVID-19 pandemic have supported household spending. Labour markets in advanced economies remain very strong, with unemployment rates in most of our key trading partners close to record low levels (figure 3.2). This has helped underpin significant growth in nominal wages in many key trading partner economies.

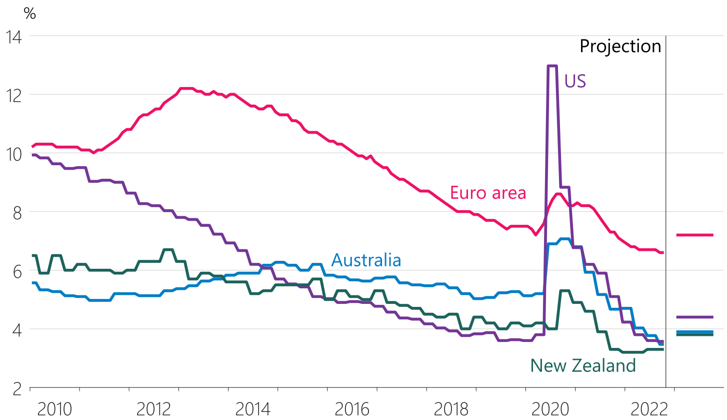
Figure 3.1
Advanced economy PMI



Source: Haver Analytics.

Note: Index values above 50 are expansionary while below 50 are contractionary.

Figure 3.2
Key trading partner unemployment rate forecasts



Source: Bloomberg, Consensus Economics.

Note: New Zealand forecasts are from Consensus Economics for comparability.

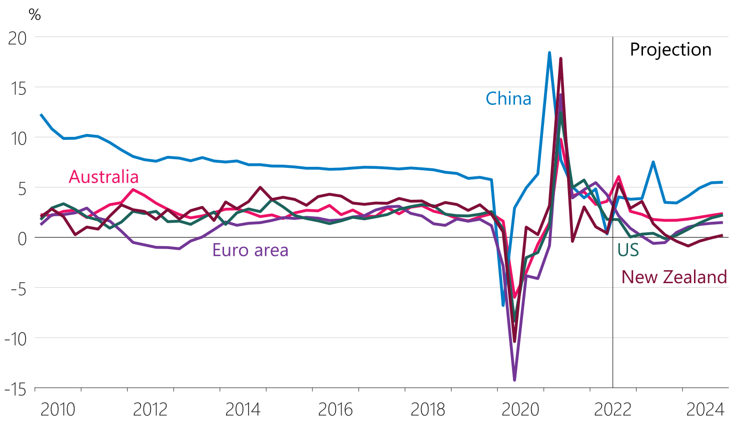
The global growth slowdown is expected to be broad based and persistent

Global growth forecasts have continued to be revised lower. Annual growth in New Zealand's trading partners as a whole is forecast to slow to 2.4 percent in 2022 and 2.0 percent in 2023, well below the average growth rate of 3.1 percent since 2000 (figure 3.3). Uncertainty around these estimates is high, with risks skewed to the downside.

China's economy is facing significant headwinds, with continued stringent COVID-19 restrictions and weakness in the property sector dampening economic confidence and activity. Annual GDP growth in China declined to 3.9 percent in the September quarter of 2022, and growth is expected to remain historically low through to at least the end of 2023.

Figure 3.3

Annual key trading partner GDP growth rate forecasts



Source: Haver Analytics, Consensus Economics, RBNZ estimates.

Note: Growth rate forecasts are based on Consensus Economics estimates, except for New Zealand's, which is based on our current projection.

Growth in the euro area and the US is expected to fall to around zero percent in 2023. Within this average outlook, some forecasters are expecting recessions in these economies. High energy prices in Europe, due to a severe reduction in natural gas supplies from Russia, are dampening economic activity significantly. Higher interest rates are expected to reduce spending in both the euro area and the US. The extent to which economic activity declines will depend partly on household spending, which has been relatively resilient, as savings built up during COVID-19 lockdowns are run down. Growth expectations are higher in Australia, where strong household consumption and continued export growth have supported economic activity.

Inflationary pressures have continued to increase since August

Global inflationary pressures remain broad based and are higher than expected at the time of the August *Statement*. While headline inflation measures have declined slightly in the US and Canada due to lower energy and transport inflation, they remain well above central bank targets (table 3.1). Core inflation measures are proving particularly resilient.

Core and headline inflation measures remain elevated in most countries as a result of common international factors and similar patterns in inflation subcomponents like food, transport, and housing. However, there are substantial differences in inflation between economies, reflecting both local economic conditions and regional factors such as exposure to the war in Ukraine (table 3.2).

Table 3.1
Central bank comparison

	Monetary policy objectives	Annual headline consumer price inflation (%)	Unemployment rate (%)	Average projected output gap in 2023 (%)
New Zealand	1–3% inflation target; employment objective	7.2	3.3	1.0
Australia	2–3% inflation target; employment objective	7.3	3.4	0.6
United States	Inflation averages 2% over time; employment objective	7.7	3.7	-0.8
United Kingdom	2% inflation target	11.1	3.6	-1.0
Euro area	2% inflation target	10.6	6.6	-0.8

Source: Respective central banks and official statistical agencies, International Monetary Fund (IMF), Stats NZ, RBNZ estimates.

Note: Output gap projections are based on IMF estimates, except for New Zealand's, which is based on our current projection. All other data reflect the respective latest outturns.

Table 3.2
Annual headline CPI inflation and subcomponents in 2022

	Australia September (%)	Euro area October (%)	NZ September (%)	UK October (%)	US October (%)
Headline CPI	7.3	10.6	7.2	11.1	7.7
Inflation ex. food & energy	6.3	5.0	6.2	5.8	6.3
Food	9.0	13.1	8.0	16.2	10.9
Services	4.1	4.3	5.1	6.3	6.7

Sources: Stats NZ, Australian Bureau of Statistics, Bureau of Labor Statistics, European Statistical Office, Haver Analytics, UK Office for National Statistics.

Note: All data is monthly except for the New Zealand CPI where only quarterly data is available.

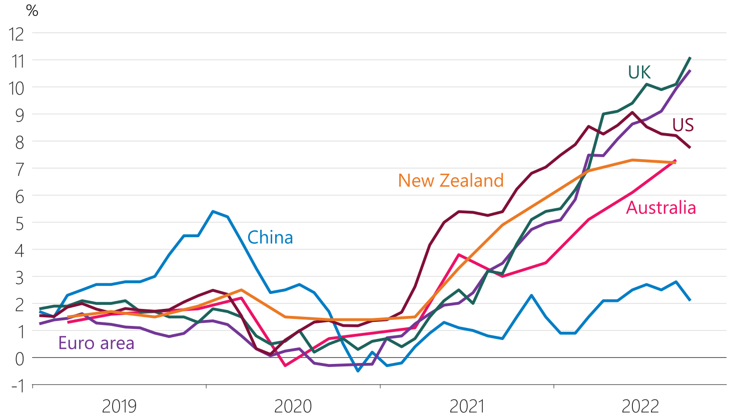
Annual headline inflation in the US remained elevated at 7.7 percent in October 2022 (figure 3.4). This reflected global inflationary pressures from food, energy and transport, as well as broader price pressures from supply-chain disruptions and stronger consumer demand. Core inflation pressures from more persistent inflation subcomponents such as shelter and services have surprised to the upside for many months (figure 3.5).

The latest US data showed some signs of price pressures moderating, with core inflation decreasing from 6.6 percent in September to 6.3 percent in October. Despite this monthly decline, there are still upside risks from factors such as stronger wage growth, supply-chain disruptions, or continued strength in shelter inflation.

Large increases in food and consumer energy prices have contributed to relatively higher inflation in Europe, reflecting the area's proximity to the war in Ukraine. Annual inflation rose in October 2022, to 10.6 percent in the Euro area, and 11.1 percent in the United Kingdom. Both headline and core inflation measures have continued to increase in Australia.

Figure 3.4

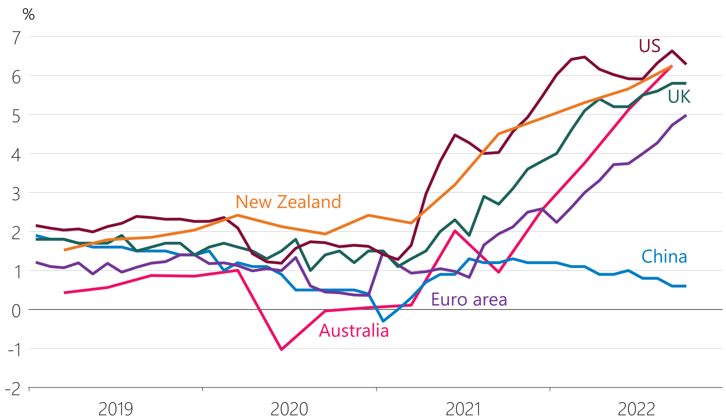
Annual headline inflation in key trading partners



Source: Haver Analytics.

Figure 3.5

Annual inflation excluding food and energy in key trading partners



Source: Haver Analytics, Organization for Economic Cooperation & Development (OECD).

Central banks have increased interest rates significantly

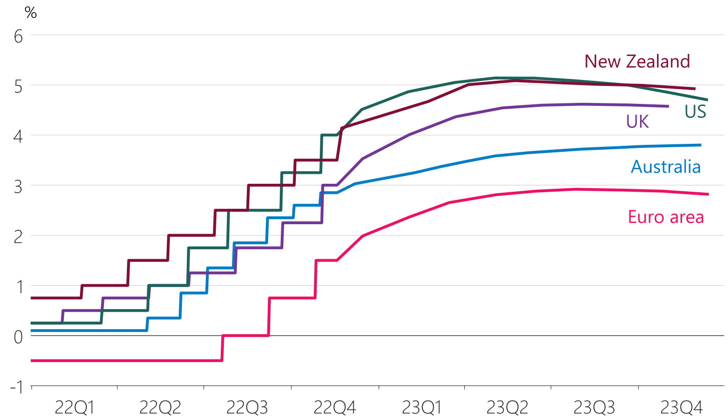
Most central banks have increased policy rates substantially in recent months, and indicated a firm commitment to maintaining tight conditions until inflationary pressures have eased. With the exception of Japan, major advanced economy central banks have each increased policy rates by a cumulative amount of between 200 and 375 basis points through 2022. Financial markets are also pricing in further policy rate increases over coming months (figure 3.6).

Market pricing for the peak in central bank policy rates during this tightening cycle has increased throughout 2022 in response to persistent inflationary pressure. However, this pricing has moderated in recent weeks, as some central banks have communicated an intent to slow the pace of monetary tightening. Reasons for a potentially slower pace of tightening include risks from the lagged impact, and the recent synchronised nature, of global monetary policy tightening. However, current market pricing for short-term interest rate increases still represents a historically fast pace.

Figure 3.6

Central bank policy interest rates and market pricing for future policy rate levels

(as at 17 Nov 2022)

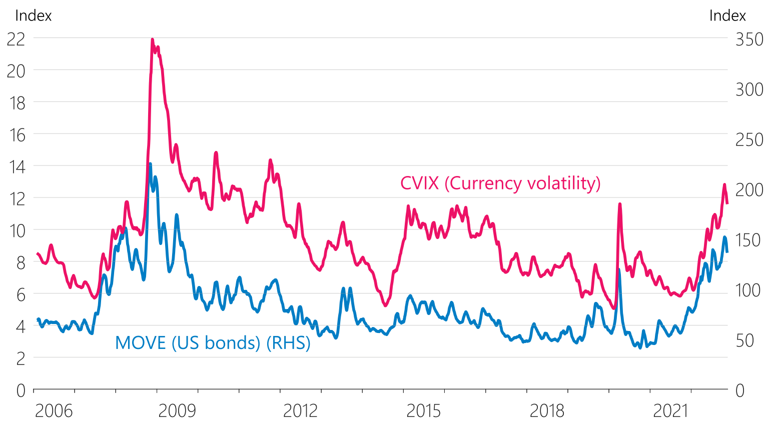


Source: Bloomberg.

Figure 3.7

Volatility indices

(monthly averages)



Source: Bloomberg.

Financial market volatility has increased

Financial markets have experienced increases in volatility over recent months as a result of greater macroeconomic uncertainty, risk aversion, and lower market liquidity. Measures of financial market volatility for bond and foreign exchange markets are above the monthly peak levels reached in March 2020 (figure 3.7).

Foreign exchange markets have been particularly volatile, as investors have sought 'safe haven' currencies such as the US dollar in response to rises in macroeconomic uncertainty. A faster pace of US monetary policy tightening has also contributed to recent US dollar strength. This is because higher US interest rates make US dollar investments relatively more attractive.

Global sovereign bond markets have also been affected by shifting monetary policy expectations. US longer-term bond yields have moved significantly higher since August, as inflation data has persistently printed above expectations. However, US yields have declined slightly through November, following signs of weaker inflationary pressures in the latest US CPI data.

Special topics

CHAPTER 04

Aoraki. Photo: Roanna McLeod

CHAPTER 4
Special topics

Before each *Statement*, the MPC is provided with analysis of some topical issues.

Topics for the November *Statement* included:

1. The neutral interest rate
2. Developments in real wages

Special topics in the past 12 months

Topic	Date/publication
Global inflation developments	MPS August 2022 (Chapter 3)
Assessing recent inflation surprises	MPS August 2022 (Chapter 3)
Business conditions	MPS August 2022 (Chapter 3)
Global inflation developments	MPS May 2022 (Chapter 4)
The outlook for household spending, house prices, and construction	MPS May 2022 (Chapter 4)
International developments and risks	MPS February 2022 (Chapter 4)
Inflation expectations	MPS February 2022 (Chapter 4)
Transmission of global developments to New Zealand's economy	MPS November 2021 (Chapter 4)
Outlook for New Zealand's productive capacity	MPS November 2021 (Chapter 4)
Climate change and inflation	MPS November 2021 (Chapter 4)

1 The neutral interest rate

The neutral interest rate is the level at which a country's policy rate is neither expansionary nor contractionary. It is conceptually the rate that, over time, would be consistent with no over- or under-utilisation of resources and stable inflation. The neutral interest rate is unobservable, but estimates of the neutral rate are useful for determining the stance of monetary policy.

The global economy heavily influences the neutral interest rate in New Zealand

In a small open economy like New Zealand's, both domestic and international developments influence the neutral interest rate. Estimates of global neutral interest rates have been declining since the 1980s. They have been driven by a variety of factors including demographic change, lower productivity growth, and changes in consumers', businesses', and governments' attitudes towards savings and investment.

New Zealand-specific factors can cause the neutral OCR to deviate from global neutral interest rates. For example, New Zealand's productivity growth has tended to be lower than those of many other advanced economies. This in part reflects an increasing proportion of low-productivity industries, and a relatively slow adoption of global technological developments into domestic production.[1] The neutral OCR can also deviate from global rates due to risk premiums placed on New Zealand assets or because of New Zealand-specific savings and investment preferences.

The Reserve Bank uses a suite of indicators that combines several models and approaches to assess the neutral interest rate. Some of the indicators are based on expectations of interest rates over the long run, or other long-run economic factors that can influence the neutral interest rate. These indicators use expectations from professional forecasters, as well as financial market participants as implied by the prices of financial instruments. Other indicators use information about recent economic outcomes and consider what they imply about the level of monetary stimulus and the neutral interest rate.

The neutral OCR can be expressed in either real – inflation-adjusted – or nominal terms. In the past, we have focused on providing estimates of the long-run nominal neutral OCR. This method was appropriate when inflation expectations were low and stable, and the economy was close to equilibrium. In the current economic environment, where inflation expectations vary considerably across different time horizons, we find it more helpful to consider real and nominal neutral interest rates separately.

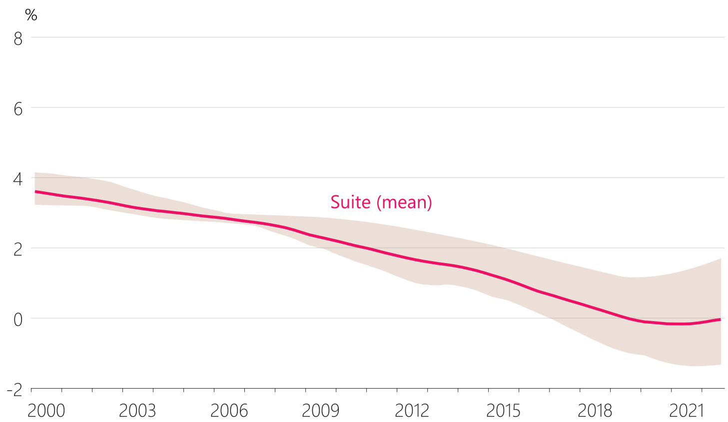
1 Productivity Commission | Productivity by the numbers (Productivity Commission, 2021).

Our average estimate of the real neutral OCR remains around 0 percent

In New Zealand, as in most economies, estimates of the real neutral interest rate have been trending downwards over several decades. In recent years, our indicator suite suggests that the real neutral interest rate has stabilised at low levels. Our current average estimate for the real neutral OCR is around 0 percent (figure 4.1). However, the wide range between the maximum and minimum values of our estimates demonstrates that there is significant uncertainty about the level of neutral interest rates, particularly since the beginning of the COVID-19 pandemic.

Figure 4.1

Real neutral interest rate indicator suite average and range



Source: RBNZ estimates.

Note: Our estimates for the neutral OCR in figures 4.1 to 4.4 end in 2022Q3 due to data availability.

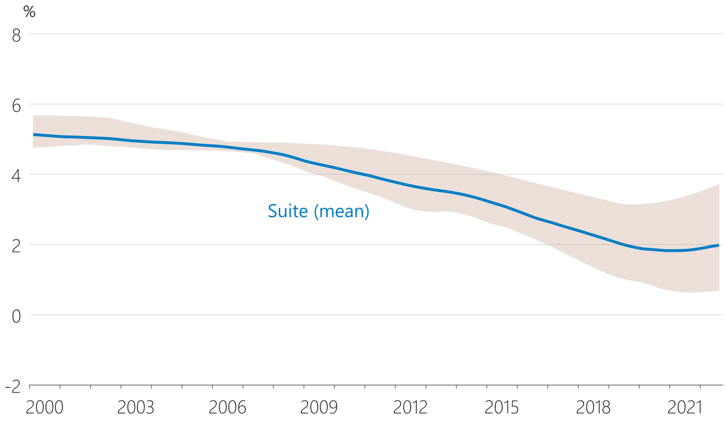
The nominal neutral OCR takes different values over different time horizons

The nominal neutral OCR consists of the real neutral OCR adjusted to account for domestic inflation expectations. Because inflation expectations differ across time horizons, the nominal neutral rate will take on different values depending on the time horizon to which it is applied.

The **long-run** nominal neutral interest rate consists of the real neutral interest rate adjusted by our 2 percent inflation target midpoint. It is the interest rate that would be neither expansionary nor contractionary when the economy is at equilibrium, with a zero output gap, employment at its maximum sustainable level, and inflation expectations anchored at our target midpoint. Our indicators suggest that the long-run nominal neutral rate has remained stable at around 2 percent (figure 4.2). However, our estimates suggest that this rate may be starting to increase slowly.

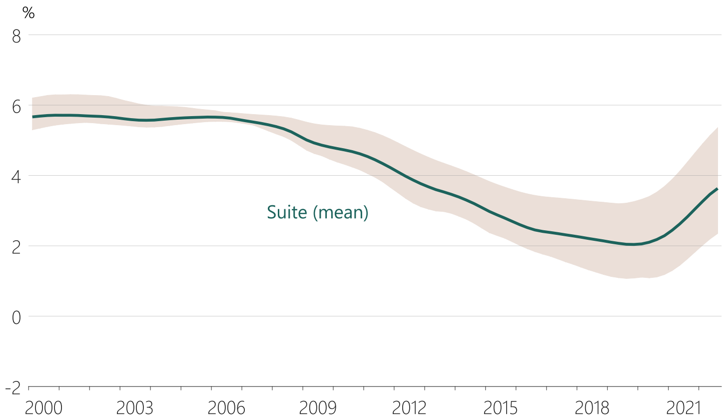
The **short-run** nominal neutral interest rate is the level of the OCR that would be neither contractionary nor stimulatory if set today. It consists of the real neutral interest rate, inflated by measures of inflation expectations that are most relevant to decision-making today. As an approximation, we use an average of several measures of inflation expectations over one- and two-year horizons. Our short-run estimates have increased relatively sharply compared to our long-run estimates, reflecting relatively rapid increases in short-run inflation expectations. The short-run nominal neutral interest rate is currently considerably higher than 2 percent (figure 4.3).

Figure 4.2
Long-run nominal neutral interest rate indicator suite average and range



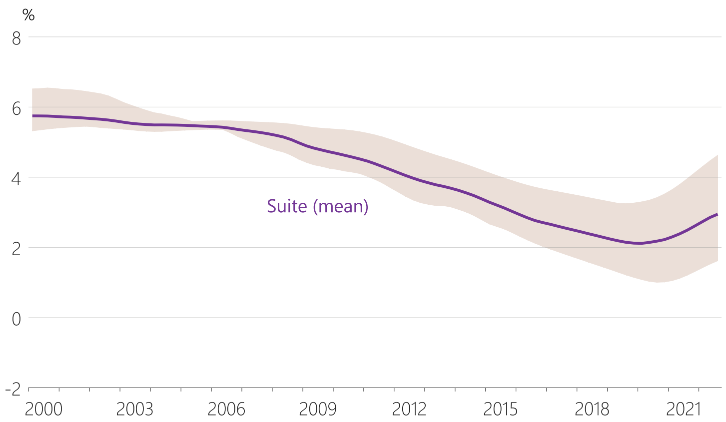
Source: RBNZ estimates.

Figure 4.3
Short-run nominal neutral interest rate indicator suite average and range



Source: RBNZ estimates.

Figure 4.4
Forecast horizon nominal neutral interest rate indicator suite average and range



Source: RBNZ estimates.

Between these two concepts is the **forecast horizon** nominal neutral OCR, which considers inflation expectations across the two- to five-year year horizon. This is the level the OCR should tend towards in order to stabilise inflation as the output gap closes. It is not always equal to the end-point of the OCR in our published projections. This is because large shocks may take more than three years to fully work through the economy, and we do not always forecast a zero output gap at the end of the projection. The forecast horizon neutral OCR sits somewhere between the short-run and long-run nominal neutral interest rate estimates (figure 4.4).

Estimates of the neutral OCR are very uncertain

There is always considerable uncertainty about the level of the neutral interest rate. This is because it is a concept that cannot be directly measured and is only imperfectly estimated. COVID-19 and the war in Ukraine have caused large structural shifts in both the global and the New Zealand economy, which further increase uncertainty. Over time we will continue to incrementally improve our estimates of the neutral interest rate and develop our understanding of recent structural shifts in the economy.

2 Developments in real wages

A tight labour market and increasing costs of living have contributed to rising nominal wages and incomes. There are a range of measures for nominal wage and income growth. Different measures are useful for informing us about changing conditions in different aspects of the labour market. These measures suggest that some workers may have been able to take advantage of the tight labour market and earn more by obtaining promotions, changing jobs or working more hours. As a result, broader wage measures generally show that earnings have roughly kept pace with CPI inflation recently.

What can different wage and income measures tell us?

Four key measures of wages and income are the Quarterly Employment Survey (QES) gross earnings per filled job, QES average hourly earnings, and the unadjusted and adjusted Labour Cost Index (LCI). These measures are described in a simple example in table 4.1.

Table 4.1

Wage and income measures example

Example	Measure
Edmund is a barista. He gets a pay rise of 6 percent.	This 6 percent is captured in the **unadjusted LCI**. This is a useful measure of wage changes for a fixed labour market composition.
Edmund's boss is asked by Stats NZ to divide this 6 percent pay rise on the basis of better quality of work (2 percent) and adjustment due to higher costs of living or meeting market rates (4 percent).	The 4 percent is captured in the **adjusted LCI**, deliberately excluding changes due to the quality of work. This is the narrowest wage inflation indicator and is a useful measure of the cost of a 'unit' of labour to a business.
Edmund's friend Kate tells him about the good wages she is earning in the construction industry and that she can get him a job earning 10 percent more than he does now. Edmund changes jobs.	This does not directly affect the above LCI measures, but affects the **QES average hourly earnings** measure. This captures changes in labour market composition (e.g. people moving to higher-paid industries) and people being promoted. It is a useful measure of labour market tightness and consumer spending potential.
In his new job, Edmund gets asked to work on weekends occasionally. So on top of earning 10 percent more per hour, he also increases his income by working longer hours.	This affects the **QES gross earnings per filled job** measure. In addition to the average hourly earnings discussed above, this captures changes in the number of hours people work. When the labour market is very tight, employers might encourage existing staff to work longer hours to cover labour shortages. Alternatively, it can be seen during periods of cost-of-living shocks where longer hours are worked by employees to maintain a given level of consumption. Being the broadest labour income measure, this is useful as an indicator of household spending potential.

Note: See Stats NZ for technical definitions of QES and LCI measures. The unadjusted LCI measure is an experimental series.

How have these different measures developed over the past few years?

Different wage measures help to inform us about different pressure points in the labour market. In the past two years, all four wage measures have increased significantly in nominal terms (figure 4.5). However, the two QES measures have increased much more than the LCI measures. This suggests that employees have increasingly used their strong positions in a very competitive labour market to increase their pay. They have gained promotions or switched jobs to better-paid positions and industries to receive higher starting wages.

From an inflation perspective, the increase in firms' unit labour cost – as measured by the adjusted LCI – has been relatively moderate compared to the overall income growth of some workers as measured by earnings per filled job. Some of this overall income growth likely represents people's increasing productivity from taking on higher-skilled jobs via promotion or job switching, and is not necessarily inflationary.

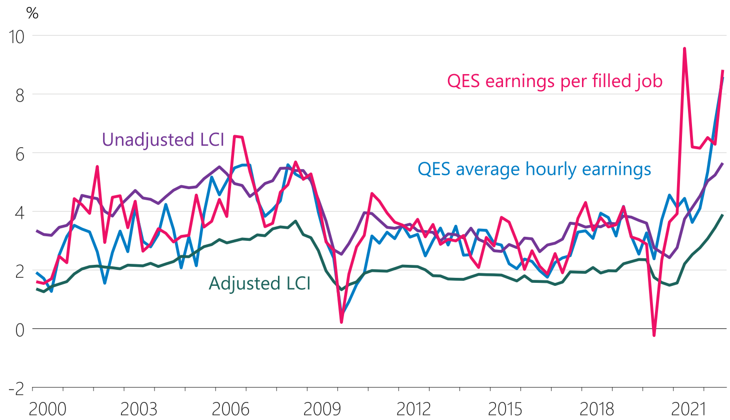
Real wages assess the quantity of goods and services that individuals' average wage incomes can purchase. Real wages rise or fall depending on whether the prices of goods and services are growing more quickly than the nominal wages used to pay for them. This determines whether an individual can afford to purchase the same amounts of these goods and services with their wages over time.

Across the four wage measures, adjusted and unadjusted LCI wages have been declining in real terms since mid-2021. However, the broader QES measures of wage and income growth have recently kept pace with CPI inflation (figure 4.6). This implies that a large part of the high wage growth some employees have experienced over this period has been needed to maintain a similar level of consumption. Some workers have been working longer hours to maintain their real incomes and standard of living.

Figure 4.5

Nominal wage and income growth

(annual)

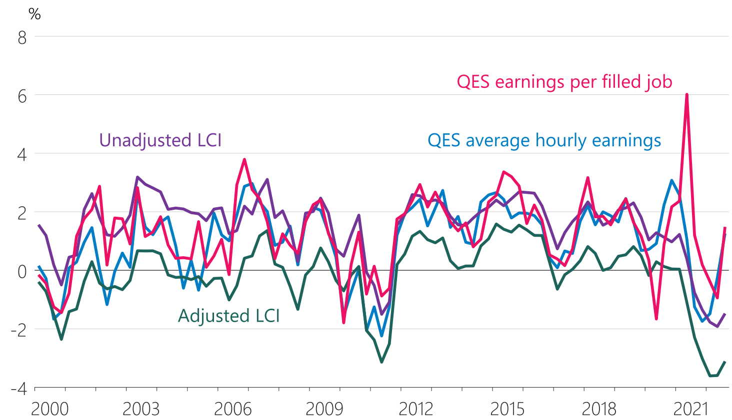


Source: Stats NZ.

Figure 4.6

Real wage and income growth

(annual)



Source: Stats NZ, RBNZ estimates.

Note: Deflated by headline CPI.

What factors drive wage growth?

Nominal wage pressures can build up when demand for labour is high and during periods of sustained high inflation. Strong labour demand encourages wage growth as employers compete for a limited pool of available workers, especially when recruiting offshore labour is difficult or expensive, such as when New Zealand's borders were closed during the COVID-19 pandemic. When unemployment is low and demand for workers is high it is easier for workers to switch jobs. This in turn incentivises firms to raise wages to retain existing staff. High CPI inflation puts pressure on employers to compensate employees for the rising cost of living, especially when there is strong competition for workers that makes it easier to move jobs.

A tight labour market often coincides with higher CPI inflation because employers pass higher wage costs into prices. We have used a simple wage Phillips curve model (figure 4.7) to help separate wage growth into the contributions from:

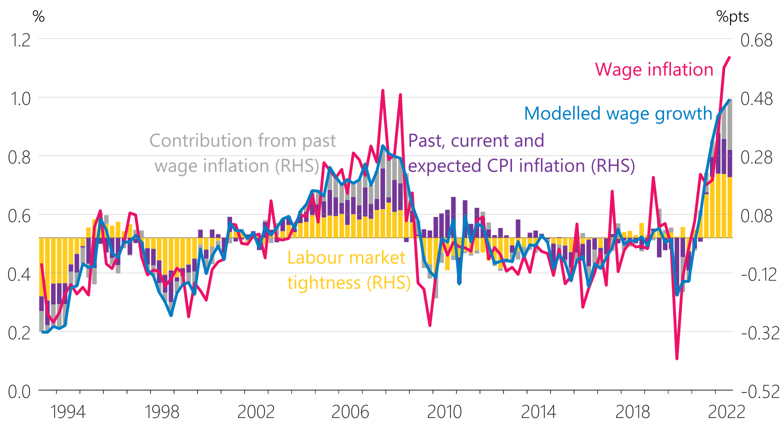
- labour market tightness;
- past, current and expected CPI inflation; and
- past wage inflation.

These factors can either work together or be offsetting forces on wage growth. A tight labour market and high CPI inflation have outweighed the economic uncertainties from COVID-19 and wage inflation has increased. In addition, wage growth tends to be persistent, meaning that recent high wage inflation is contributing more to current wage inflation.

Figure 4.7

Modelled contributions to nominal adjusted LCI wage inflation

(quarterly, seasonally adjusted)



Source: Stats NZ, RBNZ estimates.

Note: Bars show the estimated contribution of each factor to wage inflation relative to its historical average. Labour market tightness is measured by the unemployment rate gap and is lagged one quarter. Expected CPI inflation is based on businesses' expectations of CPI inflation two years ahead, as measured in the Reserve Bank's *Survey of Expectations*.

How do we expect wages to develop over the medium term?

A rise in nominal wage growth typically lags behind an increase in inflation. In most industries, wages are reviewed less often than prices and people have to become aware of rising inflation before negotiating higher pay to compensate for it. This lag means that real wages tend to first fall when inflation increases sharply, then catch up later as employees regain spending power through nominal wage increases.

Slower-moving measures of nominal wages like the LCI normally take longer to reflect such a catch-up. This is because workers tend to achieve higher pay through promotions or changing jobs first. For this reason broader measures like the QES hourly earnings have returned to positive real wage growth much more quickly in the current period of higher CPI inflation.

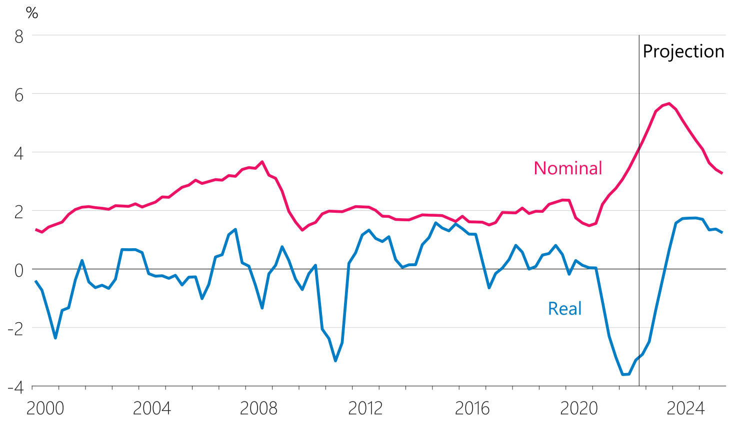
We assume that it will take until late 2023 for annual nominal adjusted LCI wage inflation to peak at around 5.7 percent. In real terms, we expect positive annual wage growth as measured by the adjusted LCI by the end of 2023 (figure 4.8).

As inflation and economic growth slow in line with rising interest rates, demand for labour will ease in the coming years. Higher net migration will also provide some relief to the current labour shortages. On this basis, we expect that nominal wage growth will eventually return to long-run average levels towards the end of the projection horizon.

Figure 4.8

Nominal and real adjusted LCI wage inflation

(annual)



Source: Stats NZ, RBNZ estimates.

Financial conditions

CHAPTER 05

Godley Head. Photo: Katie Davis

CHAPTER 5

Financial conditions

A key way in which monetary policy affects economic activity is by influencing financial conditions in New Zealand. These conditions include the interest rates at which households and businesses save and borrow, the exchange rate, and other factors such as credit availability.

Financial conditions have continued to tighten since the August *Statement*, although retail rates have increased at a slower pace than wholesale rates.

Domestic wholesale interest rates have risen significantly since the August *Statement*. Stronger global and domestic inflation data have driven market participants' expectations for the OCR over the next 12 months higher (figure 5.1). The two-year swap rate has increased by around 100 basis points since August (figure 5.2). While medium- and long-term interest rates have also risen, this increase has been smaller than for short-term interest rates. The 10-year swap rate is now 50 basis points lower than the two-year swap rate. This implies that although market participants expect higher interest rates in the near term, they expect interest rates to fall over the medium to long term.

Domestic financial market volatility remains elevated, with larger-than-normal daily interest rate changes over recent months (figure 5.3). Offshore markets have also been volatile, and developments in these markets continue to play a key role in driving changes in New Zealand interest rates. However, a number of domestic developments have also contributed to the recent pick-up in domestic interest rate volatility (table 5.1)

The New Zealand dollar has depreciated slightly on a trade-weighted basis since the August *Statement* (figure 5.4). Market risk sentiment remains weak and continues to support investor demand for the US dollar, given its status as a 'safe haven' currency. Lower prices for New Zealand's export commodities have contributed to lower demand for the New Zealand dollar. Interest rates in major economies such as the US and the euro area have increased at a faster pace than those in New Zealand, also contributing towards the recent weakness in the New Zealand dollar.

Retail interest rates continue to increase in response to increases in wholesale rates. Mortgage rates have risen across a variety of terms, with larger increases in the popular one- and two-year fixed rates, leading to a flattening of the mortgage curve (figure 5.5). Term deposit rates have also continued to increase, but remain low relative to comparable wholesale rates. This is playing a role in limiting the extent to which higher wholesale rates feed through to higher mortgage rates. We expect that higher deposit rates over the medium term will provide upward pressure to mortgage rates, independent of any changes to the OCR outlook.

Despite a recent increase in financial market volatility, domestic bank credit spreads have remained stable since the August *Statement*. Credit spreads tend to widen during periods of higher risk aversion, but they currently remain broadly in line with pre-pandemic levels (figure 5.6).

Table 5.1

Developments in domestic financial conditions as at 17 November

Wholesale interest rates	
Market pricing for the OCR	• Market participants now expect a faster and larger increase in the OCR than at the time of the August *Statement*. Financial market expectations for the peak OCR in this tightening cycle, as measured by overnight indexed swap pricing, have risen by around 120 basis points over this period to 5.1 percent. • These moves have largely been influenced by a combination of domestic and offshore economic data releases that suggest inflationary pressure is stronger than previously expected.
New Zealand Government Bonds	• New Zealand Government Bond (NZGB) yields have risen significantly since the August *Statement*. On average, yields have increased by around 80 basis points across the curve, with the 10-year trading at 4.2 percent. • NZGBs were added to the FTSE Russell World Government Bond Index (WGBI) on 1 November 2022. This led to significant purchases of NZGBs by offshore investors who use the WGBI as a benchmark. This increased demand led to NZGB yields falling relative to other benchmark domestic interest rates around the date that NZGBs were included in the index. • The NZGB market continues to function well despite a recent pick-up of volatility related to domestic and offshore factors. The intraday range for the 10 year NZGB (the difference between the lowest and highest price on a given trading day) has steadily risen since the August *Statement*. However, bid-ask spreads (how far apart buyers and sellers are from reaching an agreed price to trade) have not widened over this period. • New Zealand Debt Management issued its inaugural green sovereign bond on 15 November 2022. The bond has a maturity date of 15 May 2034 and received strong demand during the issuance process. Green bonds are structured in the same way as other nominal NZGBs; however, the funds raised are required to be put towards specific government projects with environmental benefits.
Interest rate swaps	• New Zealand dollar interest rate swaps have also increased significantly since the August *Statement*. These increases have been slightly larger at shorter terms, with the two-year swap rate increasing around 90 basis points to 4.9 percent. As a consequence, there is now a steeper inversion across the swap curve. The difference between the two- and 10-year swap rates has widened by a further 20 basis points to 50 basis points. • Global interest rate swaps have also risen sharply. These rates continue to be a key influence on domestic interest rate movements. The US two-year swap rate has risen 120 basis points since the August *Statement* to 4.7 percent. This increase largely reflects upward revisions to market expectations for the degree of further US monetary policy tightening over coming quarters.

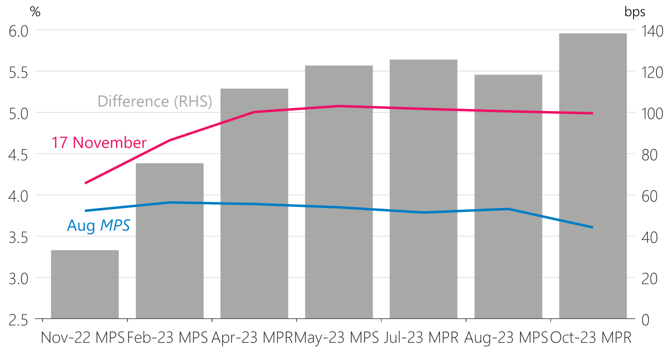
New Zealand dollar	
Interest rate differentials	• New Zealand interest rates have increased at a slower rate than interest rates in major economies such as the US and euro area since the August *Statement*. For example, over this period the NZGB two-year interest rate has increased by around 25 basis points less than the two-year US Treasury yield. Decreases in New Zealand interest rates relative to key trading partners have contributed to the depreciation in the New Zealand dollar. • However, New Zealand interest rates have risen at a faster pace than some other trading partners. This includes Australia, as the Reserve Bank of Australia has slowed down the pace of its monetary policy tightening. As a result, the New Zealand dollar has appreciated against the Australian dollar.
Risk sentiment	• During periods of elevated uncertainty or volatility, financial market participants' willingness to take on additional risk decreases and demand for 'safe haven' currencies, such as the US dollar, increases. Higher risk aversion has contributed to the depreciation of the New Zealand dollar. • Demand for riskier assets has worsened this year in response to expectations of higher interest rates, a weakening global economic outlook, and elevated uncertainty related to the economic impacts of the war in Ukraine. • These trends have continued since the August *Statement*, in particular in government bond markets. Levels of volatility have at times been similar to those observed in the initial phase of the COVID-19 pandemic in 2020.
Commodity prices	• Decreases in New Zealand's key export commodity prices have also contributed to the depreciation of the New Zealand dollar. • Prices of key export commodities in New Zealand peaked in the first quarter of this year and have gradually fallen since then. In particular, dairy prices are now near a two-year low. Despite declining over this year, commodity export prices have remained elevated relative to historical levels.
Retail interest rates	
Mortgage rates	• Mortgage rates have risen since the August *Statement* in response to rising wholesale rates. The one- and two-year fixed-rate terms have increased by 100 and 70 basis points respectively. Increases have been smaller in the three- to five-year terms, leading to a flattening of the mortgage curve. • Increases in mortgage rates continue to be smaller than increases in wholesale interest rates at comparable terms, as bank funding costs remain historically low relative to wholesale rates.
Deposit rates	• Term deposit rates have also increased since the August *Statement*. They have increased by 85 basis points and 50 basis points in the popular terms of six months and one year respectively. These increases mean that both terms are currently at their highest levels since 2015. • The deposit curve is now relatively flat, with the five-year term deposit rate currently only around 20 basis points higher than the one-year rate.

Bank funding conditions	
Funding composition	• Term deposit volumes continue to increase as depositors switch from on-call accounts. This is increasing banks' average funding costs as the interest rates paid on term deposits are higher than they are for on-call accounts. • Funding for Lending Programme (FLP) drawdowns have increased by $3.7 billion since the August *Statement*, taking total drawdowns to $16.4 billion. The additional allocation drawdown window closes on 6 December 2022. • Banks remain well funded and have strong liquidity buffers. The core funding ratio rose by 0.5 percentage points in September, to 90.6 percent. This is the highest ratio since data collection began in April 2010.
Credit spreads	• The cost of raising long-term wholesale bank funding relative to benchmark wholesale rates has been relatively stable in the second half of this year. It sits at levels higher than that in the 2020/21 period and is broadly in line with pre-pandemic levels. Short-term credit spreads have not risen and remain around the levels prevailing since the monetary response to the COVID-19 pandemic. • Domestic credit spreads have increased at a slower pace than comparable measures for offshore issuers.

Charts

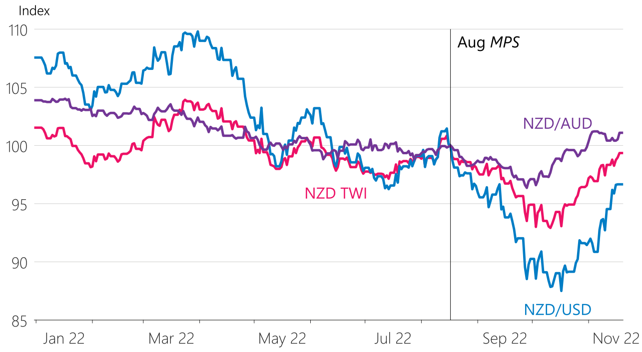
All charts are as at 17 November 2022.

Figure 5.1
Market expectations for the OCR



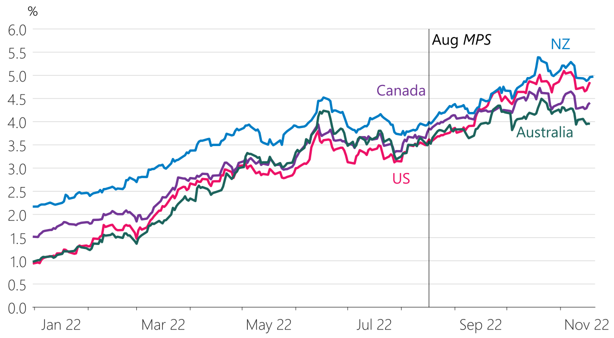
Source: Bloomberg.

Figure 5.4
New Zealand dollar TWI and selected bilateral exchange rates
(index=100 at August Statement*)*



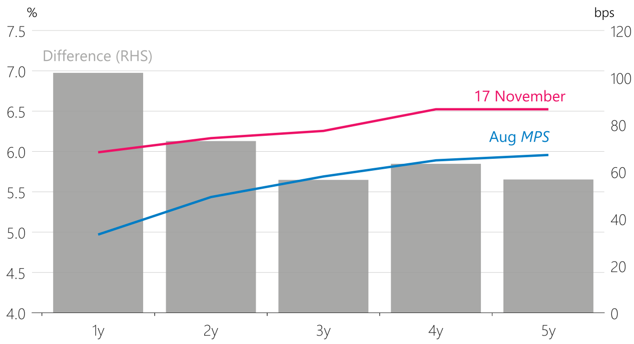
Source: RBNZ, NZFMA.

Figure 5.2
Global 2-year interest rate swaps



Source: Bloomberg.

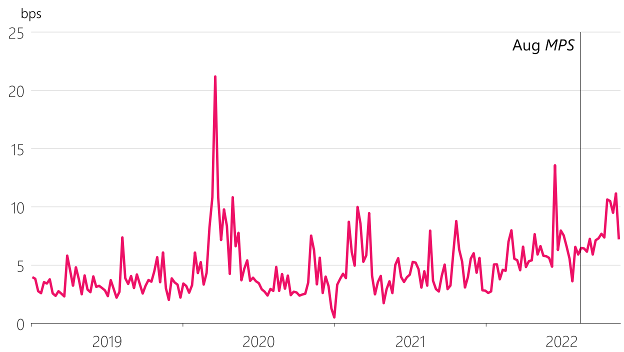
Figure 5.5
Mortgage interest rate curve



Source: interest.co.nz.

Note: The mortgage rates shown for each term are the average of the latest 'special' fixed-term rates on offer from ANZ, ASB, BNZ and Westpac.

Figure 5.3
Intraday range 10-year New Zealand Government Bond yield



Source: Bloomberg.

Figure 5.6
5-year bank credit spread

Source: Bloomberg, RBNZ estimates.

Note: This is the difference between banks' secondary market bond yields and the interest rate swap curve. This uses the averages of bonds with a remaining maturity of five years issued in New Zealand dollars by ANZ, ASB, BNZ, and Westpac. A lower spread represents a lower implied borrowing cost.

Economic projections

CHAPTER 06

Ruakākā. Photo: Liza Reiderman

CHAPTER 6
Economic projections

This chapter summarises the baseline economic projections that the MPC members considered when making their policy assessment. The projections were finalised on 17 November 2022.

These projections rely on a set of key assumptions about the global and domestic drivers of the economy. These include:

- the faster recovery of international tourism in New Zealand, which is likely to make labour shortages and inflation worse in the near term;
- a weaker exchange rate, which has supported export revenues and caused higher imported inflation;
- continued weakness in house prices and sales, which is likely to lead to an eventual slowdown in residential construction activity and household consumption;
- sustained levels of government spending in coming quarters in line with *Budget* 2022 and ongoing moderate public sector pay growth;
- higher inflation expectations, leading to more persistent inflation;
- stronger global inflation in the near term, causing central banks around the world to raise interest rates further and therefore lead to a weaker outlook for global growth; and
- slowing business investment and household consumption in response to rising costs, higher interest rates and weaker global demand.

There is significant uncertainty about these assumptions, and the actual outcomes could be different if they do not hold.

The projections take into account recent data, that show inflation remaining higher than expected, and a very tight labour market with low unemployment. The factors underpinning this starting point are discussed in chapter 2.

Inflation is expected to remain elevated in the near term. The expected strong recovery of international tourism over the 2022/23 summer will put further strain on an already stretched services sector, including hotels and restaurants. Global inflationary pressures stemming from strong global demand over the past year, and the war in Ukraine, have continued to feed through to inflation in New Zealand. Additionally, domestic demand has so far remained robust through 2022, contributing to labour shortages and putting upward pressure on prices.

Over the medium term we assume that higher interest rates in New Zealand will reduce domestic demand. This will provide space for labour shortages to ease, along with gradual population growth through natural increase and net immigration. Prices of oil and other products that New Zealand imports are expected to fall.

Consumer price inflation is expected to return to its 2 percent target midpoint towards the end of the projection horizon in 2025. Given the expected slowing in domestic demand, a rising unemployment rate from exceptionally low levels is likely. In the second half of the projection horizon, employment is expected to fall below its maximum sustainable level.

Table 6.1
Key projection assumptions

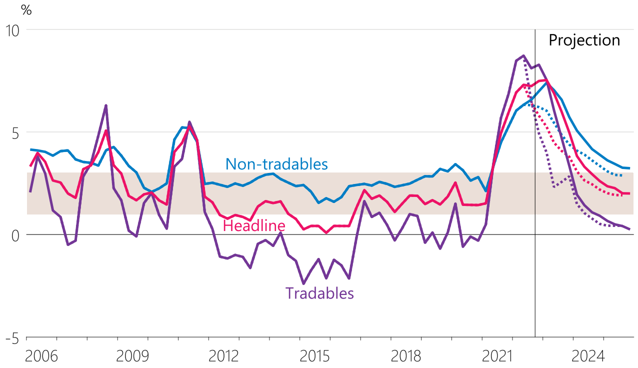
Key factors	
The international tourism restart	• The return of overseas tourists to New Zealand has exceeded expectations since the border re-opened. Forward bookings indicate demand will continue to strengthen towards a more typical pre-pandemic summer seasonal peak. We have assumed that tourism and travel services recover to 75 percent of their pre-pandemic level this summer and to 95 percent of their pre-pandemic level by the end of 2025. • It is likely that the transport, hospitality and accommodation sectors will struggle to keep up with demand over this year's peak summer period. Capacity in these sectors was reduced while the border was closed, and labour shortages remain acute. This may lead to significant price increases in those industries that operate with flexible pricing models, such as accommodation and transport.
House prices and residential construction	• House prices have fallen by 11.5 percent since their peak in November 2021. House prices are projected to keep falling into 2024, consistent with lower sales volumes in recent months. In quarterly terms, house prices are expected to fall by 19 percent by early 2024. Annual house price inflation is assumed to turn positive towards the end of the forecast horizon as net immigration increases and interest rates decline. • Higher interest rates and lower house prices are expected to reduce incentives for new home building over the projection horizon. However, home building activity has so far held up, and firms indicated in our business visits that the pipeline of residential building projects is full for about another six months. In particular, near-record house building consents, more permissive planning regulations, and delays due to supply constraints continue to support construction in the near term. Over the medium term, we project residential construction's share of potential economic output to fall back to pre-pandemic levels. • Continued house price falls are expected to slow household spending growth over the projection as aggregate household wealth falls.
Global factors	• The global economic environment is highly inflationary, particularly in the near term. Central banks have responded to high global inflation by raising interest rates. Consequently, the New Zealand dollar exchange rate has weakened, which has led to high import and export price increases for New Zealanders. • Several large economies have seen significantly higher than expected inflation readings overall since the August *Statement*. Economic growth in China is projected to remain modest as the property market downturn and COVID-19 restrictions continue to weigh on activity. Longer-term Consensus forecasts for the growth of developed nations have been revised lower, as these countries are particularly exposed to more expensive food imports and higher US interest rates. • Prices for New Zealand's commodity exports have declined in recent months. Quarterly goods export prices are assumed to decline by 10 percent over the projection, in line with softening global demand. Likewise, export goods volumes are expected to grow below their long-term trend. • Softening global demand is also assumed to decrease business investment in addition to the effects of higher interest rates. This is because the global slowdown is assumed to cause New Zealand firms to delay investment, because of either uncertainty about demand or cash-flow constraints. • The New Zealand dollar trade weighted index (TWI) is assumed to remain at 70.

Economic growth	
Production	• We expect the economy is currently still growing, supported by recent high export prices, household savings built up during the COVID-19 pandemic, and government spending. The return of international visitors is expected to keep the economy growing through the March 2023 quarter. • Economic activity has been expanding faster than capacity over the past two years, leading to a large output gap and high inflation. The average annual output gap in 2022 is estimated to be 2.4 percent of potential GDP, compared to 2.0 percent in 2021. This reflects a tighter labour market, elevated capacity pressure in the goods market and more recently a recovery in services consumption. • However, these projections contain a sustained, albeit shallow, period of economic contraction. GDP is assumed to decline by 1 percent in total from its peak; however, the timing of this contraction is highly uncertain. • We expect the output gap to start closing quickly in mid-2023, as households and businesses respond to higher interest rates in the economy, dampening demand. The annual average output gap is assumed to be -2.0 percent of potential GDP at the end of the projection. • Over the medium term, the supply of goods and services is also expected to increase. We assume that supply-chain bottlenecks will continue to loosen and that labour supply will be bolstered by a return of low but positive net immigration. Overall, the combination of slower demand and rising supply is expected to result in a negative output gap for a time.
Consumption	• Consumption growth appears to have been robust during 2022 to date. We assume that consumption continues to be supported in the near term by the Government's Cost of Living Payment and savings built up during the COVID-19 pandemic. Incomes have also been buoyed by high export prices, high labour force participation, increasing wages and longer working hours. • High interest rates and lower house prices are expected to slow consumption growth over coming quarters. Consumption is expected to decline on a per capita basis to pre-pandemic levels by mid-2024.
Investment	• We assume that the outlook for business investment remains weak, relative to its trend. Factors weighing on business investment over the projection include the weakening global outlook and elevated uncertainty, higher interest rates, declining export prices and easing capacity pressures. • We have revised our outlook for residential construction from mid-2023 significantly lower than that in the August *Statement*. This reflects an assumption that the effects of higher interest rates, tighter credit conditions, and lower house prices become more evident. Despite the softening housing market, building consent issuance has remained high so far. Due to capacity, cost, and financial pressures facing the construction sector, we expect some consented projects to be postponed or cancelled.
Government	• We assume that government activity evolves consistent with the fiscal forecasts in *Budget* 2022 in May. As in the August *Statement*, we continue to assume a smoother profile for government consumption as measured in the *System of National Accounts* relative to the May *Statement*. Elevated inflation since *Budget* 2022 is assumed to reduce government consumption in real terms.
Exports and imports	• International tourist arrivals and departures have increased considerably since border restrictions began easing earlier in the year. Exports of travel services (international visitors to New Zealand) are expected to increase considerably over the upcoming summer period. This is expected to strain capacity in the transport, hospitality and accommodation sectors. • The outlook for goods exports and imports is subdued given the expected global slowdown in demand and the expected fall in consumption in New Zealand respectively. • Quarterly goods export prices are assumed to decline by 8 percent over the projection in line with softening global demand. Likewise, export goods volumes are expected to remain below levels implied by their long-term trend. • Global central bank tightening and soft commodity prices are expected to cause non-oil import prices to decline from early 2023. By the end of 2025, non-oil import prices in real world terms are assumed to decline to their pre-pandemic level.

Labour market	
Employment and wages	• The New Zealand labour market has remained exceptionally tight. The unemployment rate remained at 3.3 percent in the September 2022 quarter, despite an increase in participation to a record level. Our suite of labour market indicators continue to suggest that the economy is well above maximum sustainable employment (MSE). • Employment is starting from a very strong level, reflecting acute labour market tightness and a record-high labour force participation rate. The number of online job vacancies is near record levels. Over the medium term, employment is projected to decline in line with the expected economic contraction and fall below MSE for a time. The slowing in domestic activity over several quarters, together with gradual growth in the labour force, is expected to relieve labour shortages. • Annual private sector LCI wage inflation is expected to continue to increase, peaking at 5.7 percent in 2023. This reflects ongoing labour shortages, expected further rises in the minimum wage and rising cost-of-living adjustments by employers.
Inflation (annual)	
Headline	• Headline inflation is expected to rise to 7.5 percent for the next two quarters. • Over 2023 and 2024, headline inflation is projected to decline, returning to the top of the 1 to 3 percent target band by late 2024. • Headline inflation is expected to continue to trend down in the latter half of the projection horizon, reaching the 2 percent midpoint of the target band at the end of the projection in 2025.
Tradables	• Tradables inflation is assumed to start declining only gradually in the next two quarters. Near-term tradables inflation is assumed to be held up by a weaker exchange rate in the current quarter and the re-imposition of the fuel excise tax in the next quarter. In the near term, higher food price inflation is caused by labour shortages, domestic shipping challenges and higher global food prices. These are expected to ease during 2023. • Over the medium term, weaker global demand and a gradual easing of international supply constraints are expected to see tradables inflation decline.
Non-tradables	• Non-tradables inflation is also expected to remain high over the next two quarters. This is being driven by ongoing labour shortages, including those likely to be exacerbated by the return of international tourism. • Shortages of materials and labour in the construction industry have continued to underpin housing-related cost increases. These capacity pressures are expected to ease as interest rates rise, house prices fall and home building activity slows. Slower domestic activity and lower house prices are expected to cause non-tradables inflation to ease over coming years.
Inflation expectations	• Elevated near-term inflation expectations are assumed to persist, putting upward pressure on inflation. Over the medium term, inflation expectations are expected to moderate in line with falling headline inflation.

Charts

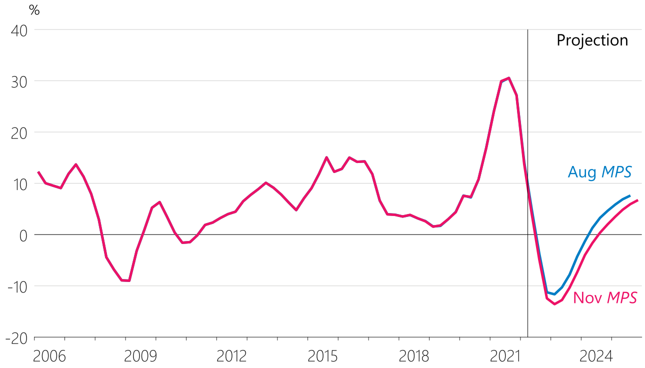
Figure 6.1
Inflation breakdown
(annual)



Source: Stats NZ, RBNZ estimates.

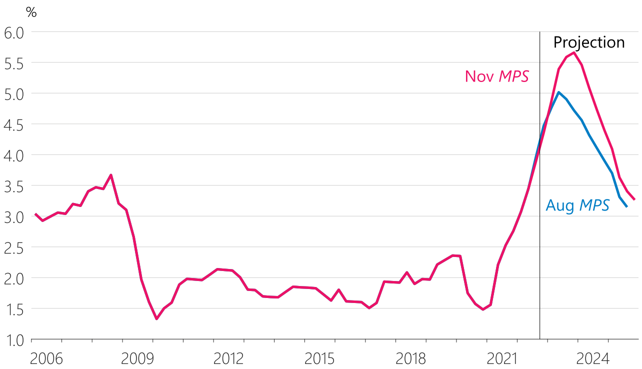
Note: Dotted lines show the projection from the August *Statement*.

Figure 6.4
House price growth
(annual)



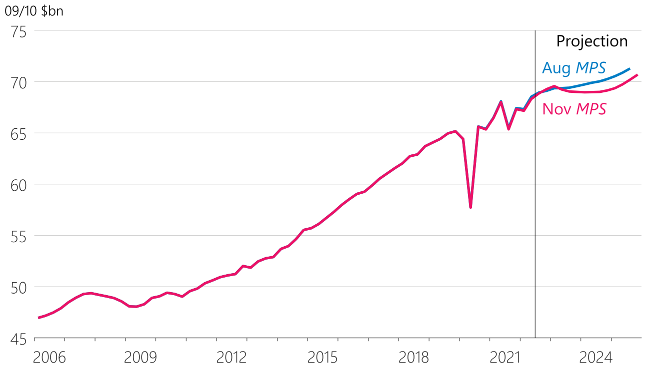
Source: CoreLogic, RBNZ estimates.

Figure 6.2
Private sector LCI wage inflation
(annual)



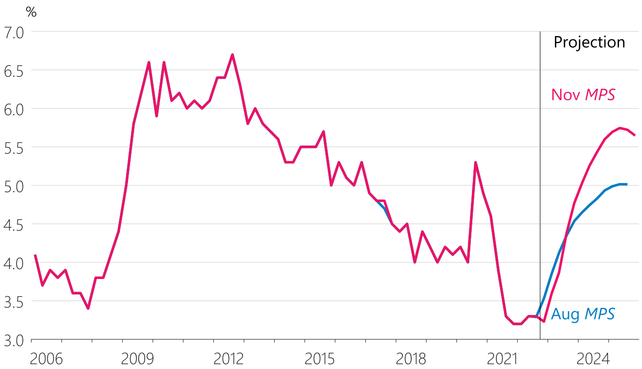
Source: Stats NZ, RBNZ estimates.

Figure 6.5
Production GDP
(quarterly, seasonally adjusted)



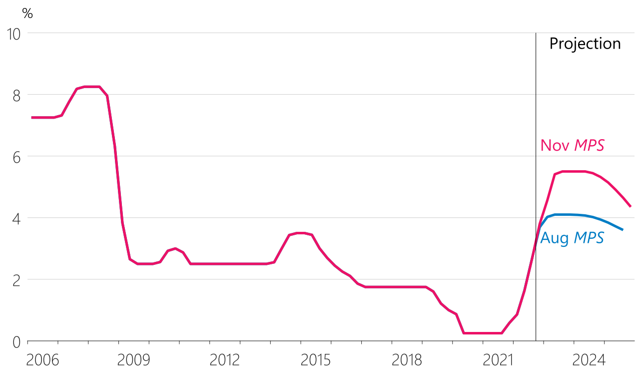
Source: Stats NZ, RBNZ estimates.

Figure 6.3
Unemployment rate
(unemployed as share of labour force, seasonally adjusted)



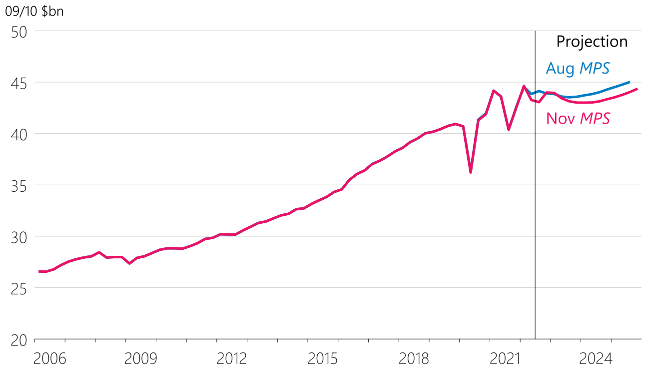
Source: Stats NZ, RBNZ estimates.

Figure 6.6
OCR
(quarterly average)



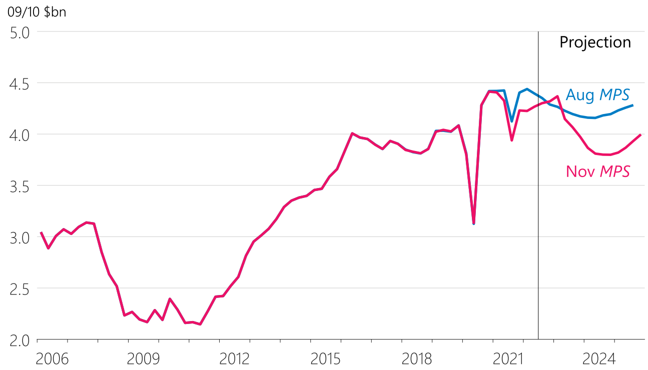
Source: RBNZ estimates.

Figure 6.7
Private consumption
(quarterly, seasonally adjusted)



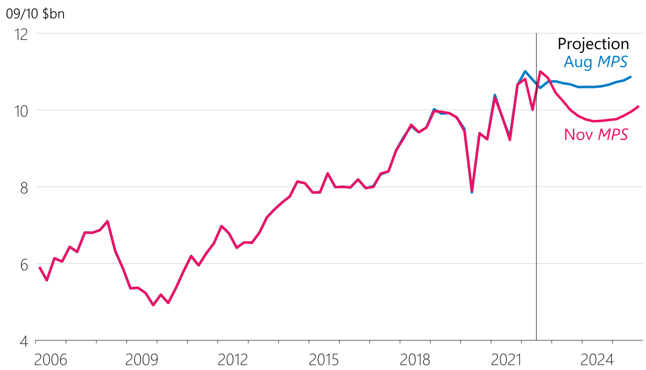
Source: Stats NZ, RBNZ estimates.

Figure 6.8
Residential investment
(quarterly, seasonally adjusted)



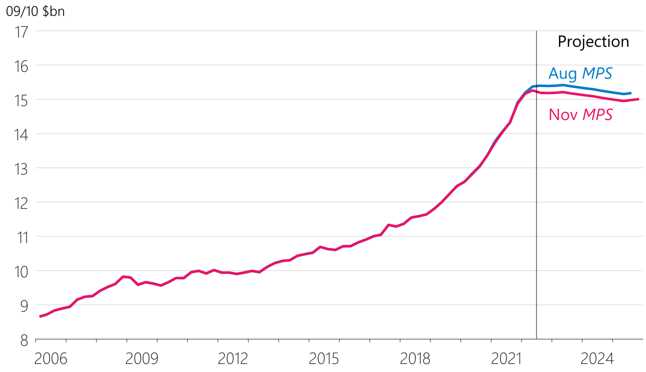
Source: Stats NZ, RBNZ estimates.

Figure 6.9
Business investment
(quarterly, seasonally adjusted)



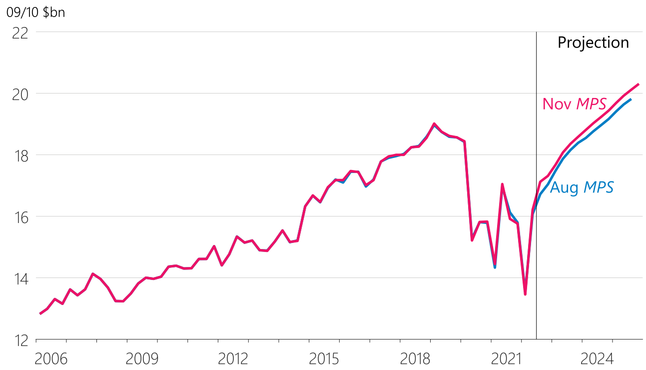
Source: Stats NZ, RBNZ estimates.

Figure 6.10
Government consumption
(quarterly, seasonally adjusted)



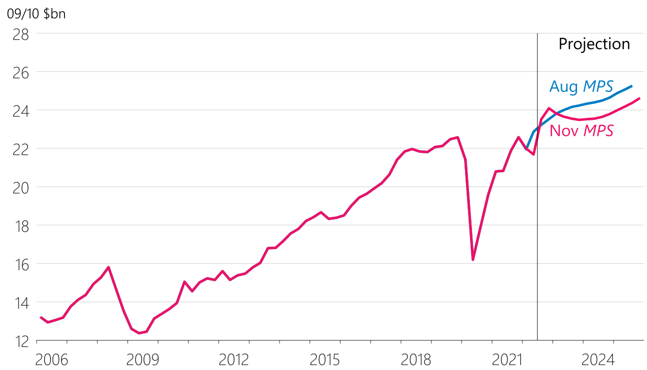
Source: Stats NZ, RBNZ estimates.

Figure 6.11
Total exports
(quarterly, seasonally adjusted)



Source: Stats NZ, RBNZ estimates.

Figure 6.12
Total imports
(quarterly, seasonally adjusted)



Source: Stats NZ, RBNZ estimates.

Appendices

CHAPTER 07

Red Crater, Tongariro National Park. Photo: Nele McHattie

CHAPTER 7
Appendices

Appendix 1: Statistical tables

Table 7.1
Key forecast variables

		GDP growth Quarterly	CPI inflation Quarterly	CPI inflation Annual	Unemployment rate	TWI	OCR
2020	Mar	-1.1	0.8	2.5	4.2	70.9	0.9
	Jun	-10.4	-0.5	1.5	4.0	69.7	0.3
	Sep	13.7	0.7	1.4	5.3	72.0	0.3
	Dec	-0.4	0.5	1.4	4.9	72.9	0.3
2021	Mar	1.7	0.8	1.5	4.6	74.9	0.3
	Jun	2.3	1.3	3.3	3.9	74.7	0.3
	Sep	-3.9	2.2	4.9	3.3	74.4	0.3
	Dec	3.0	1.4	5.9	3.2	74.3	0.6
2022	Mar	-0.2	1.8	6.9	3.2	72.6	0.9
	Jun	1.7	1.7	7.3	3.3	72.1	1.6
	Sep	**0.8**	2.2	7.2	3.3	70.6	2.7
	Dec	**0.6**	**1.7**	**7.5**	**3.2**	**70.0**	**3.8**
2023	Mar	**0.4**	**1.8**	**7.5**	**3.6**	**70.0**	**4.6**
	Jun	**-0.5**	**1.1**	**6.9**	**3.9**	**70.0**	**5.4**
	Sep	**-0.3**	**1.3**	**6.0**	**4.4**	**70.0**	**5.5**
	Dec	**-0.1**	**0.7**	**5.0**	**4.8**	**70.0**	**5.5**
2024	Mar	**-0.1**	**0.7**	**3.8**	**5.0**	**70.0**	**5.5**
	Jun	**0.0**	**0.5**	**3.3**	**5.3**	**70.0**	**5.5**
	Sep	**0.0**	**0.9**	**2.9**	**5.4**	**70.0**	**5.4**
	Dec	**0.2**	**0.4**	**2.6**	**5.6**	**70.0**	**5.3**
2025	Mar	**0.3**	**0.5**	**2.4**	**5.7**	**70.0**	**5.1**
	Jun	**0.5**	**0.5**	**2.3**	**5.7**	**70.0**	**4.9**
	Sep	**0.7**	**0.7**	**2.0**	**5.7**	**70.0**	**4.6**
	Dec	**0.7**	**0.4**	**2.0**	**5.6**	**70.0**	**4.4**

Table 7.2

Measures of inflation, inflation expectations, and asset prices

	2021				2022			
	Mar	Jun	Sep	Dec	Mar	Jun	Sep	Dec
Inflation (annual rates)								
CPI	1.5	3.3	4.9	5.9	6.9	7.3	7.2	
CPI non-tradables	2.1	3.3	4.5	5.3	6.0	6.3	6.6	
CPI tradables	0.5	3.4	5.7	6.9	8.5	8.7	8.1	
Sectoral factor model estimate of core inflation	2.6	3.2	3.8	4.3	4.8	5.2	5.4	
CPI trimmed mean (30 percent measure)	1.7	3.0	4.0	4.3	5.2	5.8	6.4	
CPI weighted median	2.3	3.0	3.3	3.8	3.9	4.8	5.0	
GDP deflator (expenditure)	0.4	2.2	3.8	5.1	6.2	5.9		
Inflation expectations								
ANZ Business Outlook – inflation 1 year ahead (quarterly average to date)	1.9	2.2	2.9	4.0	5.4	6.0	6.1	6.1
RBNZ Survey of Expectations – inflation 2 years ahead	1.9	2.0	2.3	3.0	3.3	3.3	3.1	3.6
RBNZ Survey of Expectations – inflation 5 years ahead	2.0	2.1	2.0	2.2	2.3	2.4	2.3	2.4
RBNZ Survey of Expectations – inflation 10 years ahead	2.0	2.0	2.0	2.0	2.1	2.1	2.1	2.2
Long-run inflation expectations*	2.1	2.1	2.1	2.1	2.2	2.2	2.2	2.3
Asset prices (annual percent changes)								
Quarterly house price index (CoreLogic NZ)	24.0	29.9	30.5	27.2	13.8			
REINZ Farm Price Index (quarterly average)	6.9	6.3	15.1	20.5	25.6	31.0	8.2	
NZX 50 (quarterly average to date)	14.4	17.1	10.3	3.0	-3.9	-9.0	-11.3	-14.1

Source: ANZ, Consensus Economics, RBNZ estimates.

*Long-run expectations are extracted from a range of surveys using a Nelson-Siegel model.

Table 7.3

Measures of labour market conditions

(seasonally adjusted, changes expressed in annual percent terms, unless specified otherwise)

	2021				2022		
	Mar	Jun	Sep	Dec	Mar	Jun	Sep
Household Labour Force Survey							
Unemployment rate	4.6	3.9	3.3	3.2	3.2	3.3	3.3
Underutilisation rate	12.1	10.5	9.2	9.2	9.3	9.2	9.0
Labour force participation rate	70.4	70.6	71.1	71.0	70.9	70.9	71.7
Employment rate (percentage of working-age population)	67.1	67.8	68.7	68.8	68.6	68.5	69.3
Employment growth	0.1	1.5	3.9	3.3	2.6	1.4	1.2
Average weekly hours worked	33.9	34.0	31.2	33.5	33.4	33.7	33.6
Number unemployed (thousand people)	133	114	97	93	94	96	97
Number employed (million people)	2.75	2.78	2.82	2.82	2.82	2.82	2.85
Labour force (million people)	2.88	2.89	2.92	2.91	2.91	2.91	2.95
Extended labour force (million people)	2.99	2.98	3.00	2.99	3.00	3.00	3.03
Working-age population (million people, age 15 years+)	4.09	4.10	4.10	4.10	4.11	4.11	4.12
Quarterly Employment Survey – QES							
Filled jobs growth	0.5	2.5	4.0	4.1	4.2	1.5	1.5
Average hourly earnings growth (private sector, ordinary time)	4.1	4.4	3.6	4.1	5.3	7.0	8.6
Other data sources							
Labour cost index growth, private sector, adjusted	1.6	2.2	2.5	2.8	3.1	3.4	3.9
Labour cost index growth, private sector, unadjusted	2.8	3.7	4.1	4.5	5.0	5.2	5.6
Estimated net migration (published, thousands, quarterly)	-3.5	-4.8	-3.3	-3.4			
Change in All Vacancies Index	26.7	163.7	52.1	34.5	19.3	3.4	10.3

Note: The All Vacancies Index is produced by MBIE as part of the monthly Jobs Online report, which shows changes in job vacancies advertised by businesses on internet job boards. The unadjusted LCI is an analytical index that reflects quality change in addition to price change (whereas the official LCI measures price changes only). For definitions of underutilisation, the extended labour force and related concepts, see Statistics New Zealand (2016), *Introducing underutilisation in the labour market*. Estimated net migration (published) is the Stats NZ outcomes-based measure.

Table 7.4

Composition of real GDP growth

(annual average percent change, seasonally adjusted, March years, unless specified otherwise)

	Actuals								Projection		
March year	**2015**	**2016**	**2017**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**	**2025**
Final consumption expenditure											
Private	3.3	4.2	6.4	4.8	4.5	2.5	0.5	4.6	**1.8**	**-1.0**	**0.3**
Public authority	2.9	2.1	1.9	3.7	3.4	5.8	7.5	10.3	**4.1**	**-0.3**	**-0.9**
Total	**3.2**	**3.7**	**5.4**	**4.5**	**4.3**	**3.2**	**2.1**	**6.0**	**2.4**	**-0.8**	**0.0**
Gross fixed capital formation											
Residential	8.3	7.1	8.8	-1.8	-0.1	2.8	1.8	3.0	**3.2**	**-7.0**	**-5.1**
Other	8.0	2.5	-0.1	10.3	7.0	2.5	-6.8	9.6	**5.3**	**-2.9**	**-3.3**
Total	**8.1**	**3.6**	**2.2**	**6.9**	**5.2**	**2.5**	**-4.7**	**7.9**	**4.8**	**-3.9**	**-3.7**
Final domestic expenditure	4.3	3.7	4.6	5.1	4.5	3.1	0.5	6.4	**3.0**	**-1.5**	**-0.9**
Stockbuilding*	0.5	-0.3	0.1	0.2	-0.2	-0.2	-0.2	0.7	**-0.3**	**-0.1**	**0.0**
Gross national expenditure	**4.6**	**3.2**	**4.8**	**5.6**	**4.3**	**2.8**	**-0.7**	**7.9**	**2.8**	**-1.5**	**-0.9**
Exports of goods and services	4.7	6.9	2.0	3.8	3.3	0.4	-17.6	1.4	**9.9**	**8.1**	**4.8**
Imports of goods and services	7.7	2.6	5.6	7.8	4.3	1.0	-15.9	17.2	**6.7**	**1.2**	**0.8**
Expenditure on GDP	**3.6**	**4.5**	**3.7**	**4.3**	**4.0**	**2.7**	**-0.3**	**3.7**	**3.2**	**0.0**	**0.0**
GDP (production)	3.8	3.8	3.8	3.6	3.4	2.2	-1.5	5.0	**3.0**	**0.1**	**0.1**
GDP (production, March qtr to March qtr)	3.8	4.1	3.3	3.6	3.3	0.5	3.2	1.0	**3.6**	**-0.9**	**0.6**

*Percentage point contribution to the growth rate of GDP.

Table 7.5

Summary of economic projections

(annual percent change, March years, unless specified otherwise)

	Actuals								Projection		
March year	**2015**	**2016**	**2017**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**	**2025**
Price measures											
CPI	0.3	0.4	2.2	1.1	1.5	2.5	1.5	6.9	**7.5**	**3.8**	**2.4**
Labour costs	1.8	1.8	1.5	1.9	2.0	2.4	1.6	3.1	**4.9**	**5.5**	**4.1**
Export prices (in New Zealand dollars)	-8.0	0.7	4.1	3.3	1.1	6.8	-6.2	21.4	**3.8**	**-3.1**	**-0.5**
Import prices (in New Zealand dollars)	-3.4	1.2	0.6	1.8	4.3	2.5	-2.5	19.0	**7.7**	**-3.1**	**-1.1**
Monetary conditions											
OCR (year average)	3.4	2.9	2.0	1.8	1.8	1.2	0.3	0.5	**3.2**	**5.5**	**5.4**
TWI (year average)	79.3	72.6	76.5	75.6	73.4	71.7	72.4	74.0	**70.7**	**70.0**	**70.0**
Output											
GDP (production, annual average % change)	3.8	3.8	3.8	3.6	3.4	2.2	-1.5	5.0	**3.0**	**0.1**	**0.1**
Potential output (annual average % change)	3.1	3.2	3.3	3.3	3.2	2.8	-1.3	2.8	**2.6**	**2.4**	**2.1**
Output gap (% of potential GDP, year average)	-0.7	-0.2	0.3	0.6	0.8	0.2	-0.1	2.1	**2.6**	**0.2**	**-1.8**
Labour market											
Total employment (seasonally adjusted)	3.6	2.2	5.9	2.9	1.5	2.6	0.1	2.6	**1.5**	**-0.5**	**0.4**
Unemployment rate (March qtr, seasonally adjusted)	5.5	5.3	4.9	4.4	4.2	4.2	4.6	3.2	**3.6**	**5.0**	**5.7**
Trend labour productivity	0.7	0.6	0.5	0.5	0.5	0.6	0.7	0.6	**0.5**	**0.5**	**0.5**
Key balances											
Government operating balance* (% of GDP, year to June)	0.2	0.7	1.5	1.9	2.4	-7.2	-1.4	**-5.0**	**-2.2**	**-2.0**	**-1.3**
Current account balance (% of GDP)	-3.5	-2.4	-2.6	-3.1	-3.8	-2.3	-2.5	-6.7	**-8.2**	**-7.9**	**-7.2**
Terms of trade (SNA measure, annual average % change)	0.1	-3.0	2.5	4.4	-2.5	1.9	-1.0	0.4	**-2.4**	**-1.6**	**0.2**
Household saving rate (% of disposable income)	-0.8	-0.5	0.0	-0.2	0.7	2.9	5.8	**2.5**	**1.9**	**1.3**	**1.5**
World economy											
Trading-partner GDP (annual average % change)	3.7	3.4	3.5	3.9	3.5	1.7	-0.7	5.8	**2.8**	**2.4**	**3.3**
Trading-partner CPI (TWI weighted)	1.0	1.2	1.9	1.8	1.4	2.4	0.8	4.0	**5.4**	**2.3**	**2.2**

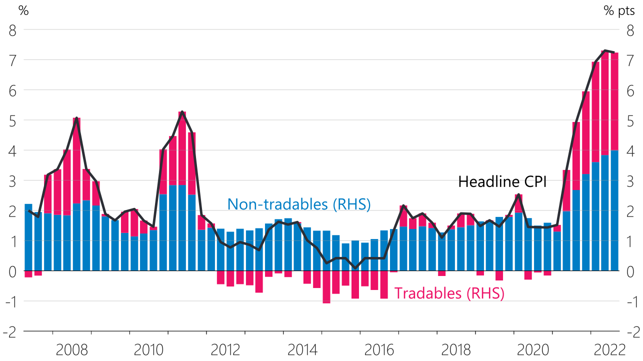
*Government operating balance is a model-based estimate of OBEGAL divided by nominal GDP in the projection. The estimate is partial because it relies on projections for some components from *Budget* 2022.

Appendix 2: Chart pack

Figure 7.1

Composition of CPI inflation

(annual)

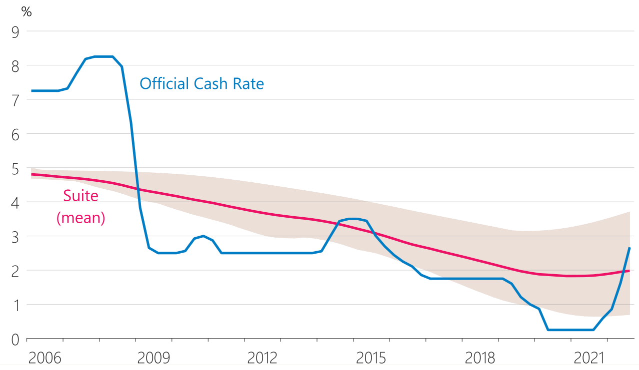


Source: Stats NZ.

Figure 7.4

OCR and neutral OCR indicator suite

(quarterly average)



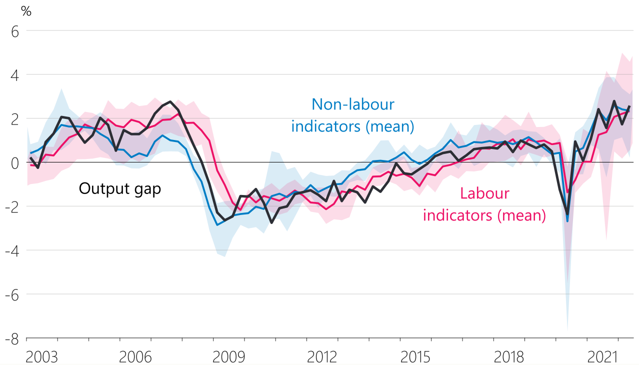
Source: RBNZ estimates.

Note: The shaded area indicates the range between the maximum and minimum values from our suite of long-run nominal neutral OCR indicators (section 4.1).

Figure 7.2

Output gap and output gap indicators

(share of potential)



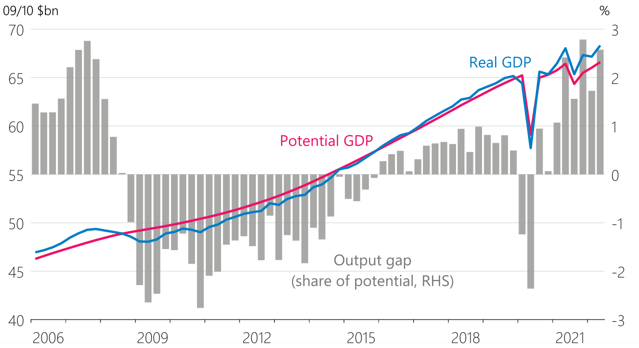
Source: NZIER, MBIE, Stats NZ, RBNZ estimates.

Note: The output gap indicators based on information about the labour market are shown separately from the other indicators. For each group of indicators, the shaded area shows the range of values and the line shows the mean value.

Figure 7.5

GDP and potential GDP

(seasonally adjusted)

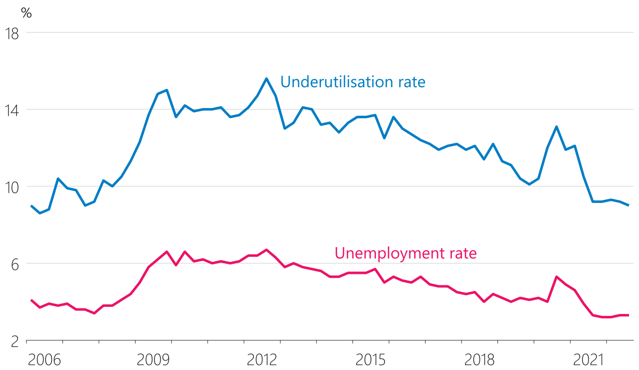


Source: Stats NZ, RBNZ estimates.

Figure 7.3

Unemployment and underutilisation rates

(seasonally adjusted)

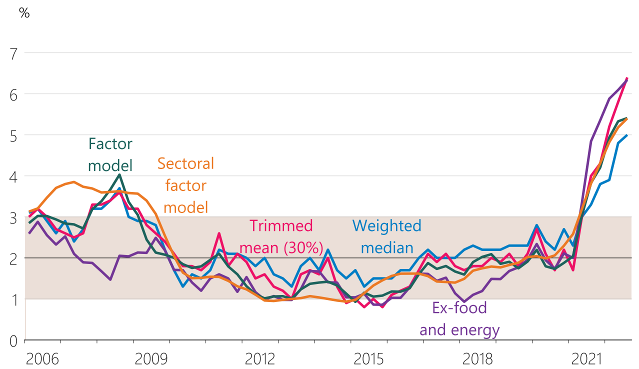


Source: Stats NZ.

Figure 7.6

Measures of core inflation

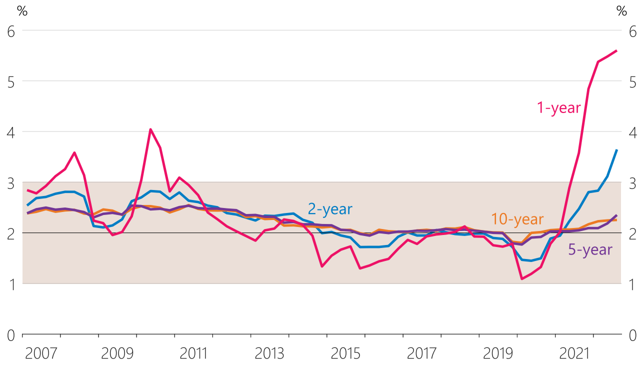
(annual)



Source: Stats NZ, RBNZ estimates.

Note: Core inflation measures exclude the GST increase in 2010.

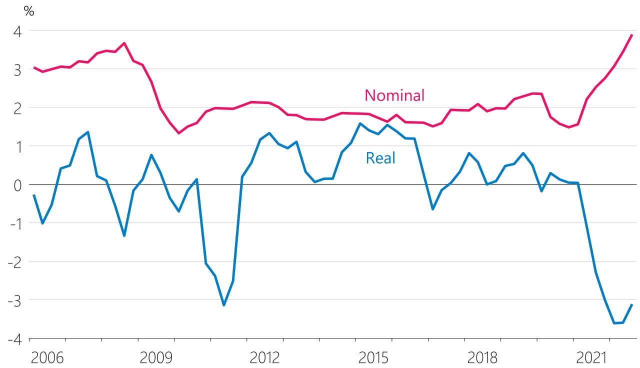
Figure 7.7
Inflation expectations
(annual)



Source: RBNZ estimates.

Note: Inflation expectations are estimates from the RBNZ inflation expectations curve, based on surveys of businesses and professional forecasters.

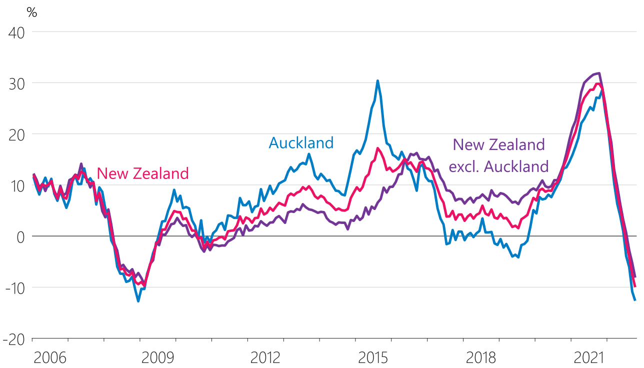
Figure 7.8
Private sector wage growth
(annual)



Source: Stats NZ, RBNZ estimates.

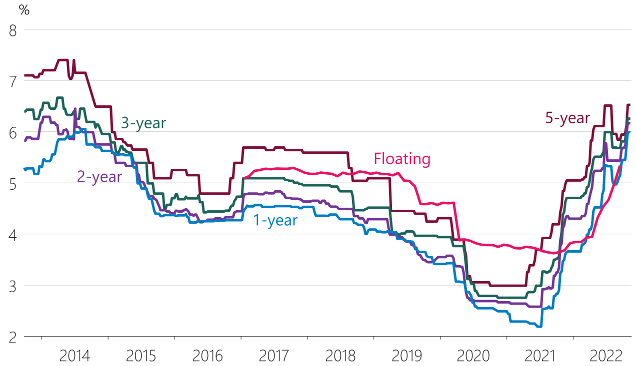
Note: Private sector wage growth is measured by the labour cost index, all salary and wage rates, private sector. Real labour cost index is deflated with headline CPI inflation.

Figure 7.9
House price inflation
(annual, nominal)



Source: REINZ.

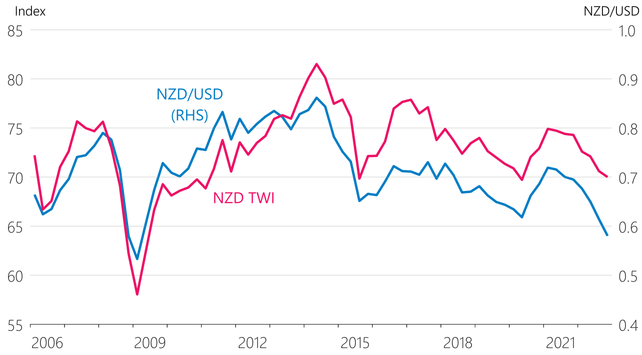
Figure 7.10
Mortgage interest rates



Source: interest.co.nz, RBNZ estimates.

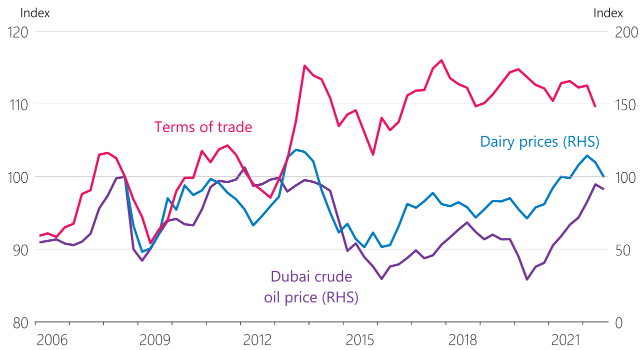
Note: The rates shown for the fixed terms are the average of the advertised rates from ANZ, ASB, BNZ, and Westpac, shown as weekly data. The floating rate represents the monthly yield on floating housing debt from the RBNZ *Income Statement survey*.

Figure 7.11
New Zealand dollar exchange rates
(quarterly average)



Source: Reuters, RBNZ.

Figure 7.12
Terms of trade, dairy and oil price indices



Source: Stats NZ, Global Dairy Trade, Reuters, RBNZ estimates.